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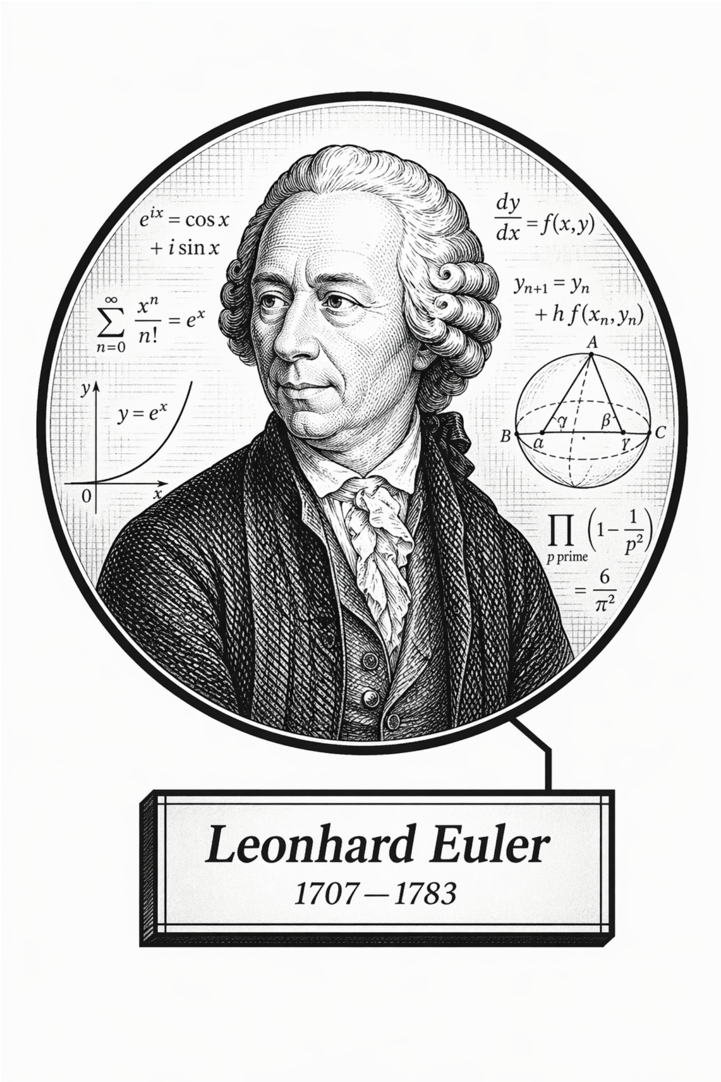
Volume IV

Algebra and Abstract Structures



I've got a number for you...

Leonhard Euler (probably)

Volume IV

Algebra and Abstract Structures

Self-Study Notes

William Sollers

June 9, 2026

Chapter 1

Introduction to Algebraic Structures

[

Where You Are in the Journey]

Propositional Logic $\to$ Predicate Calculus $\to$ Sets & Functions $\to$ Proof Techniques $\to$ Real Analysis $\to$ **Intro to Algebraic Structures** $\to$ Linear Algebra $\to$ Topology $\to$ . . .

This chapter begins with the idea of a mathematical structure and then builds toward familiar algebraic objects and number systems as examples of that idea.

Structural Roadmap

What is a structure?
$\downarrow$
Relations **Operations** **Functions** **Distinguished Elements**
$\downarrow$
How are structures assembled?
$\downarrow$
Magma $\to$ **Semigroup** $\to$ **Monoid** $\to$ **Group** $\to$ **Ring** $\to$ **Field**
$\downarrow$
How do number systems instantiate these structures?
$\downarrow$
$\mathbb{N} \to \mathbb{W} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$
$\downarrow$
How does model theory view these objects?
$\downarrow$
Structures and Models

Remark (Architectural Status). The legacy chapter has been archived intact. This live chapter is a structure-oriented skeleton only; no mathematical content has been migrated in this pass.

1.0 What Is a Mathematical Structure?

Mathematical Structures Toolkit

Structures are organized by carrier sets, operations, relations, and distinguished elements. This section fixes that vocabulary before the chapter specializes it to algebraic signatures and number-system models.

1.0.1 Structures as Mathematical Objects

Remark (Exposition). An algebraic structure is not only a set. It is a carrier set together with specified operations, relations, distinguished elements, and laws. The signature records which symbols are available; a model interprets those symbols on a concrete carrier.

1.0.2 Carrier Sets

Definition 1.0.1 (Carrier Set). The **carrier set** of a structure is the underlying set on which the structure's operations, relations, and distinguished elements are interpreted.

Remark (Standard quantified statement).

$$\mathrm{CarrierSet}(A, \mathcal{A}) \iff A \text{ is the underlying set of } \mathcal{A}.$$

Remark (Definition predicate reading). A is the carrier set of $\mathcal{A}$ exactly when every symbol of $\mathcal{A}$ is interpreted over A.

Remark (Interpretation). In a group $(G, \star)$, the carrier set is G. In a ring $(R, +, \cdot)$, the carrier set is R. The additional symbols tell us what structure has been placed on that set.

Remark (Dependencies).

- Set.

1.0.3 Operations

Remark (Exposition). Operations are function symbols interpreted as functions on the carrier set. Their arity records the number of inputs. Binary operations are central in algebra because they combine two elements of a carrier into another element of the same carrier.

1.0.4 Relations

Remark (Exposition). Relations are predicate symbols interpreted as subsets of finite Cartesian powers of the carrier set. Equality, order, divisibility, and equivalence are typical relations used to organize algebraic objects.

1.0.5 Distinguished Elements

Remark (Exposition). Distinguished elements are named elements of the carrier set. They often appear as constant symbols in a signature, such as 0 in an additive structure, 1 in a multiplicative structure, or e in a group.

1.0.6 Forward Reference: Model Theory

Remark (Exposition). Model theory separates syntax from interpretation. A signature names symbols; a structure interprets those symbols on a carrier set; axioms specify which structures count as models of a theory.

1.1 Relations

Relations Toolkit

Binary relations are predicates on ordered pairs. The structural properties used later are reflexivity, irreflexivity, symmetry, antisymmetry, asymmetry, transitivity, totality, equivalence, and order.

1.1.1 Binary Relations

Definition — Binary Relation

Definition 1.1.1 (Binary Relation). Let A be a set. A **binary relation on** A is a rule R that determines, for each ordered pair (a, b) with $a, b \in A$, whether a and b stand in the relation. We write $a\,R\,b$ when the relation holds, and $\neg(a\,R\,b)$ when it does not.

Remark (Standard quantified statement).

$$R \text{ is a binary relation on } A \iff R \subseteq A \times A.$$

Remark (Definition predicate reading). R is a binary relation on A exactly when every related pair has both coordinates in A.

Remark (Negated quantified statement).

$$\neg\,\mathrm{BinaryRelation}(R, A) \iff R \not\subseteq A \times A.$$

Remark (Interpretation). In set-theoretic language, a binary relation on A is a subset $R \subseteq A \times A$. The notation $a\,R\,b$ treats the relation as a predicate on pairs, which is the algebraic point of view used here.

Remark (Dependencies).
- Carrier set.

1.1.2 Properties of Relations

Definition 1.1.2 (Reflexive Relation). Let R be a binary relation on A. The relation R is **reflexive** if every element of A is related to itself:

$$\forall a \in A, \quad a\,R\,a.$$

Remark (Standard quantified statement).

$$\mathrm{ReflexiveRelation}(R, A) \iff R \subseteq A \times A \wedge \forall a \in A,\ a\,R\,a.$$

Remark (Definition predicate reading). R is reflexive on A exactly when every element of A has a loop to itself.

Remark (Negated quantified statement).

$$\neg\,\mathrm{ReflexiveRelation}(R, A) \iff \exists a \in A,\ \neg(a\ R\ a).$$

Remark (Interpretation). Reflexivity says that no element is excluded from relating to itself.

Remark (Dependencies).
- Binary relation.

Definition 1.1.3 (Irreflexive Relation). Let R be a binary relation on A. The relation R is **irreflexive** if no element of A is related to itself:

$$\forall a \in A, \quad \neg(a\ R\ a).$$

Remark (Standard quantified statement).

$$\mathrm{IrreflexiveRelation}(R, A) \iff R \subseteq A \times A \land \forall a \in A,\ \neg(a\ R\ a).$$

Remark (Definition predicate reading). R is irreflexive on A exactly when no element of A is related to itself.

Remark (Negated quantified statement).

$$\neg\,\mathrm{IrreflexiveRelation}(R, A) \iff \exists a \in A,\ a\ R\ a.$$

Remark (Interpretation). Irreflexivity forbids self-relation at every point of the carrier set.

Remark (Dependencies).
- Binary relation.

Definition 1.1.4 (Symmetric Relation). Let R be a binary relation on A. The relation R is **symmetric** if

$$\forall a, b \in A, \quad a\ R\ b \Rightarrow b\ R\ a.$$

Remark (Standard quantified statement).

$$\mathrm{SymmetricRelation}(R, A) \iff R \subseteq A \times A \land \forall a, b \in A,\ a\ R\ b \Rightarrow b\ R\ a.$$

Remark (Definition predicate reading). R is symmetric on A exactly when every related ordered pair can be reversed.

Remark (Negated quantified statement).

$$\neg\,\mathrm{SymmetricRelation}(R, A) \iff \exists a, b \in A,\ a\ R\ b \land \neg(b\ R\ a).$$

Remark (Interpretation). Symmetry says that the relation has no preferred direction.

Remark (Dependencies).
- Binary relation.

Definition 1.1.5 (Antisymmetric Relation). Let R be a binary relation on A. The relation R is **antisymmetric** if

$$\forall a, b \in A, \quad a\ R\ b \land b\ R\ a \Rightarrow a = b.$$

Remark (Standard quantified statement).

$$\mathrm{AntisymmetricRelation}(R, A) \iff R \subseteq A \times A \land \forall a, b \in A,\ a\ R\ b \land b\ R\ a \Rightarrow a = b.$$

Remark (Definition predicate reading). R is antisymmetric on A exactly when two-way relatedness can occur only between equal elements.

Remark (Negated quantified statement).

$$\neg\,\mathrm{AntisymmetricRelation}(R,A) \iff \exists a,b\in A,\ a\ R\ b \wedge b\ R\ a \wedge a \neq b.$$

Remark (Interpretation). Antisymmetry permits one-way comparison and self-comparison, but forbids nontrivial cycles of length two.

Remark (Dependencies).
- Binary relation.

Definition 1.1.6 (Asymmetric Relation). Let R be a binary relation on A. The relation R is **asymmetric** if

$$\forall a,b\in A,\quad a\ R\ b \Rightarrow \neg(b\ R\ a).$$

Remark (Standard quantified statement).

$$\mathrm{AsymmetricRelation}(R,A) \iff R\subseteq A\times A \wedge \forall a,b\in A,\ a\ R\ b \Rightarrow \neg(b\ R\ a).$$

Remark (Definition predicate reading). R is asymmetric on A exactly when every related ordered pair forbids its reverse.

Remark (Negated quantified statement).

$$\neg\,\mathrm{AsymmetricRelation}(R,A) \iff \exists a,b\in A,\ a\ R\ b \wedge b\ R\ a.$$

Remark (Interpretation). Asymmetry is stricter than antisymmetry and implies irreflexivity.

Remark (Dependencies).
- Binary relation.

Definition 1.1.7 (Transitive Relation). Let R be a binary relation on A. The relation R is **transitive** if related pairs can be chained:

$$\forall a,b,c\in A,\quad a\ R\ b \wedge b\ R\ c \Rightarrow a\ R\ c.$$

Remark (Standard quantified statement).

$$\mathrm{TransitiveRelation}(R,A) \iff R\subseteq A\times A \wedge \forall a,b,c\in A,\ a\ R\ b \wedge b\ R\ c \Rightarrow a\ R\ c.$$

Remark (Definition predicate reading). R is transitive on A exactly when every two-step R-chain closes to a one-step relation.

Remark (Negated quantified statement).

$$\neg\,\mathrm{TransitiveRelation}(R,A) \iff \exists a,b,c\in A,\ a\ R\ b \wedge b\ R\ c \wedge \neg(a\ R\ c).$$

Remark (Interpretation). Transitivity is the chaining law behind order, equivalence, and divisibility.

Remark (Dependencies).
- Binary relation.

Definition 1.1.8 (Total Relation). Let R be a binary relation on A. The relation R is **total** if every pair of elements is related in at least one direction:

$$\forall a,b\in A,\quad a\ R\ b \vee b\ R\ a.$$

Remark (Standard quantified statement).

$$\mathrm{TotalRelation}(R, A) \iff R \subseteq A \times A \land \forall a, b \in A,\ a\ R\ b \lor b\ R\ a.$$

Remark (Definition predicate reading). R is total on A exactly when every pair of elements is comparable by R.

Remark (Negated quantified statement).

$$\lnot\,\mathrm{TotalRelation}(R, A) \iff \exists a, b \in A,\ \lnot(a\ R\ b) \land \lnot(b\ R\ a).$$

Remark (Interpretation). Totality rules out incomparable pairs.

Remark (Dependencies).
- Binary relation.

1.1.3 Equivalence Relations

Definition — Equivalence Relation

Definition 1.1.9 (Equivalence Relation). A binary relation R on A is an **equivalence relation** if it is reflexive, symmetric, and transitive. We write $a \sim b$ when R is an equivalence relation and the particular relation is understood from context.

Remark (Standard quantified statement).

$$\begin{aligned}\mathrm{EquivalenceRelation}(R, A) \iff R &\subseteq A \times A \\ &\land [(\forall a \in A)\ a\ R\ a] \\ &\land [(\forall a, b \in A)\ a\ R\ b \Rightarrow b\ R\ a] \\ &\land [(\forall a, b, c \in A)\ a\ R\ b \land b\ R\ c \Rightarrow a\ R\ c].\end{aligned}$$

Remark (Definition predicate reading). R is an equivalence relation on A exactly when it is reflexive, symmetric, and transitive.

Remark (Negated quantified statement).

$$\lnot\,\mathrm{EquivalenceRelation}(R, A) \iff \lnot\,\mathrm{ReflexiveRelation}(R, A) \lor \lnot\,\mathrm{SymmetricRelation}(R, A) \lor \lnot\,\mathrm{Transitive}$$

Remark (Interpretation). An equivalence relation on A is precisely the data needed to partition A into disjoint, exhaustive equivalence classes. Quotient groups, quotient rings, quotient vector spaces, and quotient topologies are all built by choosing equivalence relations compatible with the relevant operations.

Remark (Dependencies).
- Binary relation.
- Reflexive relation.
- Symmetric relation.
- Transitive relation.

1.1.4 Order Relations

Definition — Partial Order

Definition 1.1.10 (Partial Order). A binary relation R on A is a **partial order** if it is reflexive, antisymmetric, and transitive. A set A equipped with a partial order R is a **partially ordered set** or **poset**, written (A, R).

Remark (Standard quantified statement).

$$\mathrm{PartialOrder}(R, A) \iff \mathrm{ReflexiveRelation}(R, A) \wedge \mathrm{AntisymmetricRelation}(R, A) \wedge \mathrm{TransitiveRelation}(R$$

Remark (Definition predicate reading). R partially orders A exactly when it supports self-comparison, antisymmetric comparison, and chaining.

Remark (Negated quantified statement).

$$\neg\,\mathrm{PartialOrder}(R, A) \iff \neg\,\mathrm{ReflexiveRelation}(R, A) \vee \neg\,\mathrm{AntisymmetricRelation}(R, A) \vee \neg\,\mathrm{TransitiveRel}$$

Remark (Interpretation). Partial orders allow incomparable elements while retaining enough structure for comparison and chaining.

Remark (Dependencies).

- Binary relation.
- Reflexive relation.
- Antisymmetric relation.
- Transitive relation.

Definition 1.1.11 (Strict Partial Order). A binary relation $<$ on A is a **strict partial order** if it is irreflexive, asymmetric, and transitive.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(<, A) \iff \mathrm{IrreflexiveRelation}(<, A) \wedge \mathrm{AsymmetricRelation}(<, A) \wedge \mathrm{TransitiveRelatio}$$

Remark (Definition predicate reading). $<$ is a strict partial order exactly when it compares without self-comparison, forbids reverse comparison, and chains transitively.

Remark (Negated quantified statement).

$$\neg\,\mathrm{StrictPartialOrder}(<, A) \iff \neg\,\mathrm{IrreflexiveRelation}(<, A) \vee \neg\,\mathrm{AsymmetricRelation}(<, A) \vee \neg\,\mathrm{Transitive}$$

Remark (Interpretation). Strict partial orders encode comparison without equality cases.

Remark (Dependencies).

- Irreflexive relation.
- Asymmetric relation.
- Transitive relation.

Definition — Total Order

Definition 1.1.12 (Total Order). A partial order $\leq$ on A is a **total order** if

$$\forall a, b \in A, \quad a \leq b \vee b \leq a.$$

A set equipped with a total order is a **totally ordered set** or **chain**.

Remark (Standard quantified statement).

$$\text{TotalOrder}(\leq, A) \iff \text{PartialOrder}(\leq, A) \wedge \forall a, b \in A,\ a \leq b \vee b \leq a.$$

Remark (Definition predicate reading). $\leq$ totally orders A exactly when it is a partial order and every pair of elements is comparable.

Remark (Negated quantified statement).

$$\neg\,\text{TotalOrder}(\leq, A) \iff \neg\,\text{PartialOrder}(\leq, A) \vee \exists a, b \in A,\ \neg(a \leq b) \wedge \neg(b \leq a).$$

Remark (Interpretation). Partial orders allow incomparable elements. Total orders require every pair to be comparable. A strict order can be recovered from a non-strict order by declaring $a < b$ exactly when $a \leq b$ and $a \neq b$.

Remark (Dependencies).
- Partial order.
- Total relation.

1.2 Operations

Operations Toolkit

Operations are functions whose outputs remain in the carrier set. Their arity records how many inputs are used: nullary operations select constants, unary operations transform one element, and binary operations combine two elements.

1.2.1 Binary Operations

Definition — Binary Operation

Definition 1.2.1 (Binary Operation). Let G be a set. A **binary operation** on G is a function

$$\star : G \times G \to G.$$

For $a, b \in G$, the element $\star(a, b)$ is denoted by $a \star b$.

Remark (Standard quantified statement).

$$\text{BinaryOperation}(\star, G) \iff \forall a \in G\, \forall b \in G,\ a \star b \in G.$$

Remark (Definition predicate reading). $\star$ is a binary operation on G exactly when it combines any two elements of G and returns an element of G.

Remark (Negated quantified statement).

$$\neg\,\mathrm{BinaryOperation}(\star, G) \iff \exists a, b \in G,\ a \star b \notin G.$$

Remark (Interpretation). A binary operation takes two elements of the carrier set and returns another element of the same set. The codomain G encodes closure.

Remark (Dependencies). • Carrier set.

1.2.2 Unary Operations

Definition — Unary Operation

Definition 1.2.2 (Unary Operation). Let S be a set. A **unary operation** on S is a function

$$f : S \to S.$$

Remark (Standard quantified statement).

$$\mathrm{UnaryOperation}(f, S) \iff f : S \to S.$$

Remark (Definition predicate reading). f is a unary operation on S exactly when it takes one element of S and returns one element of S.

Remark (Negated quantified statement).

$$\neg\,\mathrm{UnaryOperation}(f, S) \iff \exists x \in S,\ f(x) \notin S.$$

Remark (Interpretation). Unary operations appear throughout algebra: negation $a \mapsto -a$, multiplicative inverse $a \mapsto a^{-1}$, complex conjugation $z \mapsto \bar{z}$, and set complementation are all one-input operations.

Remark (Dependencies). • Function.

- Carrier set.

1.2.3 Nullary Operations

Definition 1.2.3 (Nullary Operation). Let S be a set. A **nullary operation** on S is the selection of a distinguished element of S.

Remark (Standard quantified statement).

$$\mathrm{NullaryOperation}(c, S) \iff c \in S.$$

Remark (Definition predicate reading). c is a nullary operation on S exactly when it selects a distinguished element of S.

Remark (Negated quantified statement).

$$\neg\,\mathrm{NullaryOperation}(c, S) \iff c \notin S.$$

Remark (Interpretation). In model-theoretic language, a constant symbol is interpreted as a nullary operation. The identity elements 0, 1, and e are typical algebraic examples.

Remark (Dependencies). • Carrier set.

1.3 Laws of Operations

Laws of Operations Toolkit

Operation laws are universally quantified constraints on operations. The chapter uses associativity, commutativity, distributivity, idempotence, and cancellation to separate one algebraic structure from another.

1.3.1 Associativity

Definition — Associativity

Definition 1.3.1 (Associativity). Let $\star$ be a binary operation on G. The operation $\star$ is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in G.$$

Remark (Standard quantified statement).

$$\text{Associative}(\star, G) \;:\equiv\; \forall a, b, c \in G,\ (a \star b) \star c = a \star (b \star c).$$

Remark (Definition predicate reading). $\star$ is associative on G exactly when any threefold product has the same value under either parenthesization.

Remark (Negated quantified statement).

$$\neg\,\text{Associative}(\star, G) \;:\equiv\; \exists a, b, c \in G,\ (a \star b) \star c \neq a \star (b \star c).$$

Remark (Interpretation). Associativity says that parenthesization does not affect the result of a chain of operations.

Remark (Dependencies).

- Binary operation.

1.3.2 Commutativity

Definition — Commutativity

Definition 1.3.2 (Commutativity). Let $\star$ be a binary operation on G. The operation $\star$ is **commutative** if

$$a \star b = b \star a \quad \text{for all } a, b \in G.$$

Remark (Standard quantified statement).

$$\text{Commutative}(\star, G) \;:\equiv\; \forall a, b \in G,\ a \star b = b \star a.$$

Remark (Definition predicate reading). $\star$ is commutative on G exactly when swapping the two inputs never changes the output.

Remark (Negated quantified statement).

$$\neg\,\text{Commutative}(\star, G) \;:\equiv\; \exists a, b \in G,\ a \star b \neq b \star a.$$

Remark (Interpretation). Commutativity says that the order of the two inputs does not affect the result. Associativity and commutativity are independent properties.

Remark (Dependencies).

- Binary operation.

1.3.3 Distributivity

Definition 1.3.3 (Left Distributivity). Let $+$ and $\cdot$ be binary operations on A. Multiplication is **left distributive** over addition if

$$\forall a, b, c \in A, \quad a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark (Standard quantified statement).

$$\text{LeftDistributive}(\cdot, +, A) \iff \forall a, b, c \in A,\ a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark (Definition predicate reading). Multiplication is left distributive over addition exactly when multiplying a sum on the left equals the sum of the two left products.

Remark (Negated quantified statement).

$$\neg\,\text{LeftDistributive}(\cdot, +, A) \iff \exists a, b, c \in A,\ a \cdot (b + c) \neq a \cdot b + a \cdot c.$$

Remark (Interpretation). Left distributivity lets left multiplication pass through addition.

Remark (Dependencies).

- Binary operation.

Definition 1.3.4 (Right Distributivity). Let $+$ and $\cdot$ be binary operations on A. Multiplication is **right distributive** over addition if

$$\forall a, b, c \in A, \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

Remark (Standard quantified statement).

$$\text{RightDistributive}(\cdot, +, A) \iff \forall a, b, c \in A,\ (a + b) \cdot c = a \cdot c + b \cdot c.$$

Remark (Definition predicate reading). Multiplication is right distributive over addition exactly when multiplying a sum on the right equals the sum of the two right products.

Remark (Negated quantified statement).

$$\neg\,\text{RightDistributive}(\cdot, +, A) \iff \exists a, b, c \in A,\ (a + b) \cdot c \neq a \cdot c + b \cdot c.$$

Remark (Interpretation). Right distributivity lets right multiplication pass through addition.

Remark (Dependencies).

- Binary operation.

Remark (Interpretation). Distributivity is the bridge between the additive and multiplicative operations in rings and fields.

1.3.4 Idempotence

Definition 1.3.5 (Idempotent Operation). Let $\star$ be a binary operation on A. The operation $\star$ is **idempotent** if

$$\forall a \in A, \quad a \star a = a.$$

Remark (Standard quantified statement).

$$\mathrm{IdempotentOperation}(\star, A) \iff \mathrm{BinaryOperation}(\star, A) \land \forall a \in A,\ a \star a = a.$$

Remark (Definition predicate reading). $\star$ is idempotent exactly when repeating the same input does not change it.

Remark (Negated quantified statement).

$$\neg\,\mathrm{IdempotentOperation}(\star, A) \iff \exists a \in A,\ a \star a \neq a.$$

Remark (Interpretation). Idempotence says that self-combination stabilizes immediately.

Remark (Dependencies).

- Binary operation.

Remark (Migration TODO). The archived chapter contains material on idempotence, but it was not migrated in this pass because the archive does not contain a clearly isolated idempotence discussion in the legacy Algebraic Structures chapter. Source: archive/algebra-and-abstract-structures-legacy.tex.

1.3.5 Cancellation Laws

Proposition 1.3.6 (Cancellation Laws in a Group). *Let G be a group and let $a, b, c \in G$. If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.*

Remark (Standard quantified statement).

$$\mathrm{Group}(G, \star) \implies \forall a, b, c \in G,\ (a \star b = a \star c \Rightarrow b = c) \land (b \star a = c \star a \Rightarrow b = c).$$

Remark (Theorem predicate reading). In a group, a common left or right factor can be cancelled from an equation.

Remark (Negated quantified statement).

$$\mathrm{Group}(G, \star) \land \exists a, b, c \in G,\ (a \star b = a \star c \land b \neq c) \lor (b \star a = c \star a \land b \neq c) \implies \bot.$$

Remark (Interpretation). Cancellation is the algebraic effect of having inverses.

Remark (Dependencies).

- Group.
- Inverse element.

Remark (Proof TODO). Proof exists in the archived chapter and should be migrated to a one-proof-per-file artifact. Source: archive/algebra-and-abstract-structures-legacy.tex.

1.4 Functions

Functions Toolkit

Functions are total, single-valued transformations. Injective, surjective, and bijective functions measure how faithfully a map covers its codomain, while morphisms preserve selected structure.

1.4.1 Functions as Structure-Preserving Objects

Definition — Function

Definition 1.4.1 (Function). Let A and B be sets. A **function** from A to B is a relation $f \subseteq A \times B$ such that:

(i) $\forall a \in A,\ \exists b \in B,\ (a,b) \in f$;

(ii) if $(a, b_1) \in f$ and $(a, b_2) \in f$, then $b_1 = b_2$.

We write $f : A \to B$ and $f(a) = b$ in place of $(a,b) \in f$.

Remark (Standard quantified statement).

$$\mathrm{Function}(f, A, B) \iff \forall a \in A,\ \exists!\, b \in B \text{ such that } (a,b) \in f.$$

Remark (Definition predicate reading). f is a function from A to B exactly when every input in A has exactly one output in B.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Function}(f, A, B) \iff \exists a \in A \text{ with no output in } B \vee \exists a \in A \text{ with two distinct outputs in } B$$

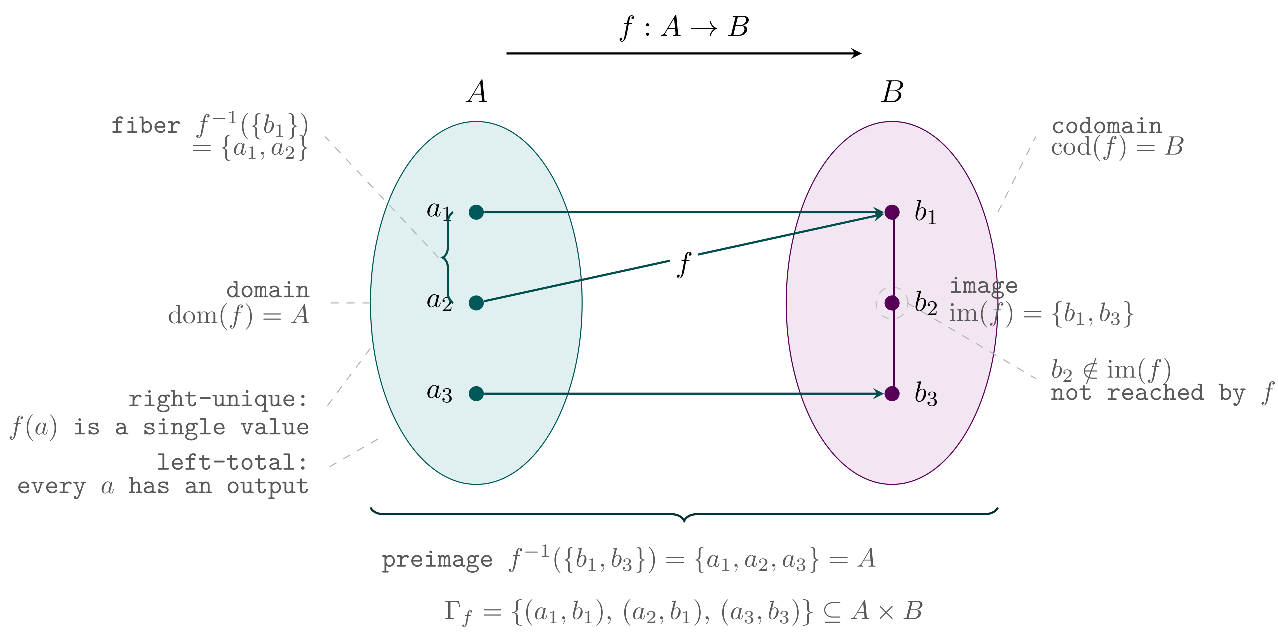


Figure 1.1: Vocabulary of a function $f : A \to B$.

Remark (Interpretation). A function assigns exactly one output in B to every input in A. In algebra, functions become structure-preserving objects when they are required to respect specified operations, relations, or constants.

Remark (Dependencies).

- Binary relation.
- Domain and codomain sets.

1.4.2 Injective Functions

Definition — Injective

Definition 1.4.2 (Injective Function). Let $f : A \to B$. The function f is **injective** if

$$\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Remark (Standard quantified statement).

$$\text{Injective}(f, A, B) \iff f : A \to B \wedge \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Remark (Definition predicate reading). f is injective exactly when equal outputs force equal inputs.

Remark (Negated quantified statement).

$$\neg\,\text{Injective}(f, A, B) \iff \exists a_1, a_2 \in A,\ a_1 \neq a_2 \wedge f(a_1) = f(a_2).$$

Remark (Interpretation). Injectivity means distinct inputs do not collide under the function.

Remark (Dependencies).
- Function.

1.4.3 Surjective Functions

Definition — Surjective

Definition 1.4.3 (Surjective Function). Let $f : A \to B$. The function f is **surjective** if

$$\forall b \in B,\ \exists a \in A,\ f(a) = b.$$

Remark (Standard quantified statement).

$$\text{Surjective}(f, A, B) \iff f : A \to B \wedge \forall b \in B,\ \exists a \in A,\ f(a) = b.$$

Remark (Definition predicate reading). f is surjective exactly when every element of the codomain is hit by some input.

Remark (Negated quantified statement).

$$\neg\,\text{Surjective}(f, A, B) \iff \exists b \in B,\ \forall a \in A,\ f(a) \neq b.$$

Remark (Interpretation). Surjectivity says that the image of f fills the whole codomain.

Remark (Dependencies).
- Function.

1.4.4 Bijective Functions

Definition — Bijective

Definition 1.4.4 (Bijective Function). Let $f : A \to B$. The function f is **bijective** if it is both injective and surjective.

Remark (Standard quantified statement).

$$\text{Bijective}(f, A, B) \iff \text{Injective}(f, A, B) \wedge \text{Surjective}(f, A, B).$$

Remark (Definition predicate reading). f is bijective exactly when it is both injective and surjective.

Remark (Negated quantified statement).

$$\neg\,\text{Bijective}(f, A, B) \iff \neg\,\text{Injective}(f, A, B) \vee \neg\,\text{Surjective}(f, A, B).$$

Remark (Interpretation). Bijectivity is the set-theoretic part of invertibility. A bijective structure-preserving map is the usual route to an isomorphism.

Remark (Dependencies).

- Injective function.
- Surjective function.

1.4.5 Morphisms

Definition — Homomorphism

Definition 1.4.5 (Homomorphism). Let A and B be algebraic structures of the same type. A **homomorphism** $f : A \to B$ is a function that preserves the operations of the structure.

Remark (Standard quantified statement).

$$\text{Homomorphism}(f, A, B) \iff f : A \to B \text{ and } f \text{ preserves the named operations of } A \text{ and } B.$$

Remark (Definition predicate reading). f is a homomorphism exactly when it is a structure-preserving function.

Remark (Negated quantified statement).

$$\neg\,\text{Homomorphism}(f, A, B)$$

when f is not a function $A \to B$ or fails to preserve at least one named operation.

Remark (Interpretation). Homomorphisms are the maps that keep algebraic operations visible through a function.

Remark (Dependencies).

- Function.
- Operations.

Definition 1.4.6 (Monomorphism). A **monomorphism** is an injective homomorphism.

Remark (Standard quantified statement).

$$\text{Monomorphism}(f, A, B) \iff \text{Homomorphism}(f, A, B) \wedge \text{Injective}(f, A, B).$$

Remark (Definition predicate reading). A monomorphism is a homomorphism that embeds without collisions.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Monomorphism}(f, A, B) \iff \neg\,\mathrm{Homomorphism}(f, A, B) \lor \neg\,\mathrm{Injective}(f, A, B).$$

Remark (Interpretation). Monomorphisms preserve structure while keeping distinct inputs distinct.

Remark (Dependencies).

- Homomorphism.
- Injective function.

Definition 1.4.7 (Epimorphism). An **epimorphism** is a surjective homomorphism.

Remark (Standard quantified statement).

$$\mathrm{Epimorphism}(f, A, B) \iff \mathrm{Homomorphism}(f, A, B) \land \mathrm{Surjective}(f, A, B).$$

Remark (Definition predicate reading). An epimorphism is a homomorphism that reaches every element of the codomain.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Epimorphism}(f, A, B) \iff \neg\,\mathrm{Homomorphism}(f, A, B) \lor \neg\,\mathrm{Surjective}(f, A, B).$$

Remark (Interpretation). Epimorphisms preserve structure while covering the whole target.

Remark (Dependencies).

- Homomorphism.
- Surjective function.

Definition 1.4.8 (Isomorphism). An **isomorphism** is a bijective homomorphism.

Remark (Standard quantified statement).

$$\mathrm{Isomorphism}(f, A, B) \iff \mathrm{Homomorphism}(f, A, B) \land \mathrm{Bijective}(f, A, B).$$

Remark (Definition predicate reading). An isomorphism is a structure-preserving bijection.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Isomorphism}(f, A, B) \iff \neg\,\mathrm{Homomorphism}(f, A, B) \lor \neg\,\mathrm{Bijective}(f, A, B).$$

Remark (Interpretation). Isomorphisms identify two structures as the same up to relabeling.

Remark (Dependencies).

- Homomorphism.
- Bijective function.

Definition 1.4.9 (Endomorphism). An **endomorphism** is a homomorphism from a structure to itself.

Remark (Standard quantified statement).

$$\mathrm{Endomorphism}(f, A) \iff \mathrm{Homomorphism}(f, A, A).$$

Remark (Definition predicate reading). An endomorphism is a structure-preserving self-map.

Remark (Negated quantified statement).

$$\neg\,\text{Endomorphism}(f, A) \iff \neg\,\text{Homomorphism}(f, A, A).$$

Remark (Interpretation). Endomorphisms are internal symmetries or transformations, not necessarily invertible ones.

Remark (Dependencies).
- Homomorphism.

Definition 1.4.10 (Automorphism). An **automorphism** is a bijective endomorphism.

Remark (Standard quantified statement).

$$\text{Automorphism}(f, A) \iff \text{Endomorphism}(f, A) \wedge \text{Bijective}(f, A, A).$$

Remark (Definition predicate reading). An automorphism is an invertible structure-preserving self-map.

Remark (Negated quantified statement).

$$\neg\,\text{Automorphism}(f, A) \iff \neg\,\text{Endomorphism}(f, A) \vee \neg\,\text{Bijective}(f, A, A).$$

Remark (Interpretation). Automorphisms are the reversible symmetries of a structure.

Remark (Dependencies).
- Endomorphism.
- Bijective function.

1.5 Distinguished Elements

Distinguished Elements Toolkit

Distinguished elements are named constants or elements selected by equations: identities do nothing, inverses undo, absorbing elements collapse an operation, and constants appear as part of a signature.

1.5.1 Identity Elements

Definition 1.5.1 (Identity Element). Let $\star$ be a binary operation on A. An element $e \in A$ is an **identity element** for $\star$ if

$$\forall a \in A, \quad e \star a = a \star e = a.$$

Remark (Standard quantified statement).

$$\text{IdentityElement}(e, \star, A) \iff e \in A \wedge \forall a \in A,\ e \star a = a \star e = a.$$

Remark (Definition predicate reading). e is an identity for $\star$ exactly when multiplying by e on either side leaves every element unchanged.

Remark (Negated quantified statement).

$$\neg\,\text{IdentityElement}(e, \star, A) \iff e \notin A \vee \exists a \in A,\ e \star a \neq a \vee a \star e \neq a.$$

Remark (Interpretation). An identity element is the distinguished element that does nothing under the operation.

Remark (Dependencies).
- Binary operation.

Proposition 1.5.2 (Uniqueness of the Identity). *In a group, the identity element is unique.*

Remark (Standard quantified statement).

$$\mathrm{Group}(G, \star) \implies \exists! e \in G \ \mathrm{IdentityElement}(e, \star, G).$$

Remark (Theorem predicate reading). Every group has exactly one identity element.

Remark (Negated quantified statement).

$$\mathrm{Group}(G, \star) \wedge \exists e_1, e_2 \in G, \ \mathrm{IdentityElement}(e_1, \star, G) \wedge \mathrm{IdentityElement}(e_2, \star, G) \wedge e_1 \neq e_2 \implies \perp.$$

Remark (Interpretation). The symbol for the identity is not extra arbitrary data once the group operation is fixed.

Remark (Dependencies).
- Group.
- Identity element.

Remark (Proof TODO). Proof exists in the archived chapter and should be migrated to a one-proof-per-file artifact. Source: archive/algebra-and-abstract-structures-legacy.tex.

1.5.2 Inverses

Definition 1.5.3 (Inverse Element). Let $(G, \star)$ be a group with identity element e. An element $b \in G$ is an **inverse** of $a \in G$ if

$$a \star b = b \star a = e.$$

Remark (Standard quantified statement).

$$\mathrm{InverseElement}(b, a, \star, e, G) \iff a, b, e \in G \wedge a \star b = b \star a = e.$$

Remark (Definition predicate reading). b is an inverse of a exactly when composing a and b in either order returns the identity.

Remark (Negated quantified statement).

$$\neg \mathrm{InverseElement}(b, a, \star, e, G) \iff b \notin G \vee a \star b \neq e \vee b \star a \neq e.$$

Remark (Interpretation). An inverse is an element that undoes another element with respect to the group operation.

Remark (Dependencies).
- Group.
- Identity element.

Proposition 1.5.4 (Uniqueness of Inverses). *In a group, every element has a unique inverse.*

Remark (Standard quantified statement).

$$\operatorname{Group}(G,\star) \implies \forall a \in G,\ \exists! b \in G\ \operatorname{InverseElement}(b,a,\star,e,G).$$

Remark (Theorem predicate reading). Every element of a group has exactly one inverse.

Remark (Negated quantified statement).

$$\operatorname{Group}(G,\star) \wedge \exists a,b,c \in G,\ \operatorname{InverseElement}(b,a,\star,e,G) \wedge \operatorname{InverseElement}(c,a,\star,e,G) \wedge b \neq c \implies \bot.$$

Remark (Interpretation). Once the group operation and identity are fixed, inverse notation is well-defined.

Remark (Dependencies).

- Group.
- Inverse element.
- Identity element.

Remark (Proof TODO). Proof exists in the archived chapter and should be migrated to a one-proof-per-file artifact. Source: archive/algebra-and-abstract-structures-legacy.tex.

1.5.3 Absorbing Elements

Definition 1.5.5 (Absorbing Element). Let $\star$ be a binary operation on A. An element $z \in A$ is an **absorbing element** for $\star$ if

$$\forall a \in A, \quad z \star a = a \star z = z.$$

Remark (Standard quantified statement).

$$\operatorname{AbsorbingElement}(z,\star,A) \iff z \in A \wedge \forall a \in A,\ z \star a = a \star z = z.$$

Remark (Definition predicate reading). z is absorbing for $\star$ exactly when combining any element with z on either side returns z.

Remark (Negated quantified statement).

$$\neg \operatorname{AbsorbingElement}(z,\star,A) \iff z \notin A \vee \exists a \in A,\ z \star a \neq z \vee a \star z \neq z.$$

Remark (Interpretation). An absorbing element collapses every product involving it back to itself.

Remark (Dependencies).

- Binary operation.

Remark (Examples). The element 0 is absorbing for multiplication in the integers, reals, and complex numbers.

1.5.4 Distinguished Constants

Remark (Exposition). Distinguished constants are the model-theoretic presentation of named elements. The group identity e, additive identity 0, and multiplicative identity 1 are constants when they are part of the signature.

1.6 Algebraic Structures

Algebraic Structures Toolkit

An algebraic structure is a carrier set equipped with operations and laws. The live hierarchy here is magma, semigroup, monoid, group, abelian group, ring, integral domain, and field.

1.6.1 Structure Signatures

Definition — First-Order Signature

Definition 1.6.1 (First-Order Signature). A **first-order signature** $\mathcal{L}$ consists of constant symbols, function symbols with specified arities, and relation symbols with specified arities.

Remark (Standard quantified statement).

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

where each function and relation symbol has a specified arity.

Remark (Definition predicate reading). A first-order signature is the syntactic list of symbols that a structure must interpret.

Remark (Interpretation). The signature describes what kinds of constants, operations, and relations are visible before any particular carrier set is chosen.

Remark (Dependencies).

- Constants.
- Operations.
- Relations.

Definition — $\mathcal{L}$-Structure

Definition 1.6.2 ($\mathcal{L}$-Structure). Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** $\mathcal{A}$ consists of a nonempty carrier set A together with interpretations of each constant, function, and relation symbol of $\mathcal{L}$ on A.

Remark (Standard quantified statement).

$$\mathcal{A} \models \mathcal{L} \iff \mathcal{A} = (A, \dots) \text{ interprets each symbol of } \mathcal{L} \text{ on } A.$$

Remark (Definition predicate reading). $\mathcal{A}$ is an $\mathcal{L}$-structure exactly when it gives meanings to the symbols of $\mathcal{L}$ on a carrier set.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Structure}_{\mathcal{L}}(\mathcal{A})$$

when some symbol of $\mathcal{L}$ is not interpreted with the required arity on the carrier set.

Remark (Interpretation). The signature is the type; the structure is the instance. A group signature names multiplication, identity, and inverse, while a particular group interprets those symbols on a carrier set.

Remark (Dependencies).

- First-order signature.
- Carrier set.

1.6.2 Magmas

Definition — Magma

Definition 1.6.3 (Magma). A **magma** is a pair $(A, \star)$ where A is a nonempty set and $\star : A \times A \to A$ is a binary operation.

Remark (Standard quantified statement).

$$\mathrm{Magma}(A, \star) \iff A \neq \varnothing \wedge \mathrm{BinaryOperation}(\star, A).$$

Remark (Definition predicate reading). $(A, \star)$ is a magma exactly when A is nonempty and $\star$ is closed on A.

Remark (Negated quantified statement).

$$\neg\, \mathrm{Magma}(A, \star) \iff A = \varnothing \vee \neg\, \mathrm{BinaryOperation}(\star, A).$$

Remark (Interpretation). A magma is the weakest nontrivial algebraic structure: it has a carrier set and a closed binary operation, with no associativity, identity, inverse, or commutativity requirements.

Remark (Dependencies).

- Carrier set.
- Binary operation.

1.6.3 Semigroups

Definition — Semigroup

Definition 1.6.4 (Semigroup). A **semigroup** is a pair $(A, \star)$ where A is a nonempty set and $\star : A \times A \to A$ is an associative binary operation:

$$\forall a, b, c \in A, \quad (a \star b) \star c = a \star (b \star c).$$

Remark (Standard quantified statement).

$$\mathrm{Semigroup}(A, \star) \iff \mathrm{Magma}(A, \star) \wedge \mathrm{Associative}(\star, A).$$

Remark (Definition predicate reading). $(A, \star)$ is a semigroup exactly when it is a magma whose operation is associative.

Remark (Negated quantified statement).

$$\neg\, \mathrm{Semigroup}(A, \star) \iff \neg\, \mathrm{Magma}(A, \star) \vee \neg\, \mathrm{Associative}(\star, A).$$

Remark (Interpretation). Semigroups are magmas where finite products can be written without specifying parentheses.

Remark (Dependencies).

- Magma.
- Associativity.

Remark (Examples). Nonempty strings under concatenation and positive integers under addition are standard semigroups.

1.6.4 Monoids

Definition — Monoid

Definition 1.6.5 (Monoid). A **monoid** is a triple $(A, \star, e)$ where $(A, \star)$ is a semigroup and $e \in A$ is an identity element:

$$\forall a \in A, \quad a \star e = e \star a = a.$$

Remark (Standard quantified statement).

$$\mathrm{Monoid}(A, \star, e) \iff \mathrm{Semigroup}(A, \star) \wedge \mathrm{IdentityElement}(e, \star, A).$$

Remark (Definition predicate reading). $(A, \star, e)$ is a monoid exactly when $(A, \star)$ is a semigroup and e is a two-sided identity for $\star$.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Monoid}(A, \star, e) \iff \neg\,\mathrm{Semigroup}(A, \star) \vee \neg\,\mathrm{IdentityElement}(e, \star, A).$$

Remark (Interpretation). Monoids add a do-nothing element to associative composition.

Remark (Dependencies).

- Semigroup.
- Identity element.

Remark (Examples). The natural numbers under addition with identity 0, the natural numbers under multiplication with identity 1, and strings under concatenation with the empty string are monoids.

1.6.5 Groups

Definition — Group

Definition 1.6.6 (Group). A **group** is a pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G satisfying:

G1. associativity;

G2. existence of an identity element $e \in G$;

G3. existence, for each $a \in G$, of an inverse element $a^{-1} \in G$.

Remark (Standard quantified statement).

$$\mathrm{Group}(G, \star) \iff \exists e \in G,\ \mathrm{Monoid}(G, \star, e) \wedge \forall a \in G,\ \exists b \in G\ \mathrm{InverseElement}(b, a, \star, e, G).$$

Remark (Definition predicate reading). $(G, \star)$ is a group exactly when it is an associative closed operation with an identity and an inverse for every element.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Group}(G, \star) \iff \neg\,\mathrm{Associative}(\star, G) \vee \texttt{no two-sided identity exists} \vee \texttt{some element has no two}$$

Remark (Interpretation). Groups are algebraic structures of reversible composition.

Remark (Dependencies).

- Binary operation.
- Associativity.
- Identity element.
- Inverse element.

Definition 1.6.7 (Order of a Group). The **order** of a group G, denoted $|G|$, is the cardinality of its underlying set. If $|G|$ is finite, G is a finite group; otherwise it is an infinite group.

Remark (Standard quantified statement).

$$\mathrm{OrderOfGroup}(G) = |G|.$$

Remark (Definition predicate reading). The order of G is the number of elements in the carrier set of G.

Remark (Interpretation). Finite groups are classified by a natural-number order; infinite groups have infinite cardinal order.

Remark (Dependencies).

- Group.
- Cardinality.

Remark (Examples). Examples include $(\mathbb{Z}, +)$, $(\mathbb{Q}\backslash\{0\}, \cdot)$, $(\mathbb{Z}/n\mathbb{Z}, +)$, and $GL_n(\mathbb{R})$ under matrix multiplication.

1.6.6 Abelian Groups

Definition — Abelian Group

Definition 1.6.8 (Abelian Group). A group $(G, \star)$ is **abelian** if its operation is commutative:

$$\forall a, b \in G, \quad a \star b = b \star a.$$

Remark (Standard quantified statement).

$$\mathrm{AbelianGroup}(G, \star) \iff \mathrm{Group}(G, \star) \wedge \mathrm{Commutative}(\star, G).$$

Remark (Definition predicate reading). $(G, \star)$ is abelian exactly when it is a group whose operation commutes.

Remark (Negated quantified statement).

$$\neg\,\mathrm{AbelianGroup}(G, \star) \iff \neg\,\mathrm{Group}(G, \star) \vee \exists a, b \in G,\ a \star b \neq b \star a.$$

Remark (Interpretation). Abelian groups are usually written additively: the operation is $+$, the identity is 0, and the inverse of a is $-a$.

Remark (Dependencies).
- Group.
- Commutativity.

1.6.7 Rings

Definition — Ring

Definition 1.6.9 (Ring). A **ring** is a triple $(R, +, \cdot)$ where R is a set and $+$ and $\cdot$ are binary operations on R satisfying:

R1. $(R, +)$ is an abelian group;

R2. multiplication is associative;

R3. multiplication distributes over addition on the left and on the right.

Remark (Standard quantified statement).

$$\mathrm{Ring}(R, +, \cdot) \iff \mathrm{AbelianGroup}(R, +) \wedge \mathrm{Associative}(\cdot, R) \wedge \mathrm{LeftDistributive}(\cdot, +, R) \wedge \mathrm{RightDistributive}($$

Remark (Definition predicate reading). $(R, +, \cdot)$ is a ring exactly when addition forms an abelian group, multiplication is associative, and multiplication distributes over addition.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Ring}(R, +, \cdot) \iff \neg\,\mathrm{AbelianGroup}(R, +) \vee \neg\,\mathrm{Associative}(\cdot, R) \vee \neg\,\mathrm{LeftDistributive}(\cdot, +, R) \vee \neg\,\mathrm{RightDis}$$

Remark (Interpretation). A ring is the algebraic setting where addition, subtraction, and multiplication coexist coherently.

Remark (Dependencies).
- Abelian group.
- Associativity.
- Distributivity.

Definition 1.6.10 (Commutative Ring). A ring $(R, +, \cdot)$ is **commutative** if

$$\forall a, b \in R, \quad a \cdot b = b \cdot a.$$

Remark (Standard quantified statement).

$$\mathrm{CommutativeRing}(R, +, \cdot) \iff \mathrm{Ring}(R, +, \cdot) \wedge \mathrm{Commutative}(\cdot, R).$$

Remark (Definition predicate reading). A ring is commutative exactly when its multiplication commutes.

Remark (Negated quantified statement).

$$\neg\,\mathrm{CommutativeRing}(R, +, \cdot) \iff \neg\,\mathrm{Ring}(R, +, \cdot) \vee \exists a, b \in R,\ a \cdot b \neq b \cdot a.$$

Remark (Interpretation). Commutative rings are rings whose multiplication behaves symmetrically in its two inputs.

Remark (Dependencies).

- Ring.
- Commutativity.

Definition 1.6.11 (Ring with Unity). A ring $(R, +, \cdot)$ is a **ring with unity** if there exists $1 \in R$ such that

$$\forall a \in R, \quad 1 \cdot a = a \cdot 1 = a.$$

Remark (Standard quantified statement).

$$\text{RingWithUnity}(R, +, \cdot, 1) \iff \text{Ring}(R, +, \cdot) \wedge \text{IdentityElement}(1, \cdot, R).$$

Remark (Definition predicate reading). A ring has unity exactly when multiplication has a two-sided identity.

Remark (Negated quantified statement).

$$\neg\,\text{RingWithUnity}(R, +, \cdot, 1) \iff \neg\,\text{Ring}(R, +, \cdot) \vee \neg\,\text{IdentityElement}(1, \cdot, R).$$

Remark (Interpretation). The unity element is the multiplicative identity of the ring.

Remark (Dependencies).

- Ring.
- Identity element.

Remark (Examples). The integers, residue rings $\mathbb{Z}/n\mathbb{Z}$, polynomial rings, and matrix rings are standard examples of rings.

1.6.8 Integral Domains

Definition 1.6.12 (Zero Divisor). Let R be a ring. A nonzero element $a \in R$ is a **zero divisor** if there exists a nonzero element $b \in R$ such that $ab = 0$.

Remark (Standard quantified statement).

$$\text{ZeroDivisor}(a, R) \iff a \in R \wedge a \neq 0 \wedge \exists b \in R,\ b \neq 0 \wedge ab = 0.$$

Remark (Definition predicate reading). a is a zero divisor exactly when a nonzero multiplication by a can produce zero.

Remark (Negated quantified statement).

$$\neg\,\text{ZeroDivisor}(a, R) \iff a = 0 \vee \forall b \in R,\ ab = 0 \Rightarrow b = 0.$$

Remark (Interpretation). Zero divisors identify where multiplication loses cancellation behavior.

Remark (Dependencies).

- Ring.
- Multiplicative zero.

Definition — Integral Domain

Definition 1.6.13 (Integral Domain). An **integral domain** is a commutative ring with unity in which there are no zero divisors.

Remark (Standard quantified statement).

$$\mathrm{IntegralDomain}(R,+,\cdot,0,1) \iff \mathrm{CommutativeRing}(R,+,\cdot)\wedge\mathrm{RingWithUnity}(R,+,\cdot,1)\wedge\forall a,b\in R,\ ab$$

Remark (Definition predicate reading). An integral domain is a commutative ring with 1 where a product is zero only because at least one factor is zero.

Remark (Negated quantified statement).

$$\neg\,\mathrm{IntegralDomain}(R,+,\cdot,0,1) \iff \neg\,\mathrm{CommutativeRing}(R,+,\cdot)\vee\neg\,\mathrm{RingWithUnity}(R,+,\cdot,1)\vee\exists a,b\in$$

Remark (Interpretation). Integral domains are the commutative rings where multiplicati has no nonzero zero-divisor obstruction.

Remark (Dependencies).

- Commutative ring.
- Ring with unity.
- Zero divisor.

Proposition 1.6.14 (Cancellation in Integral Domains). *Let R be an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$.*

Remark (Standard quantified statement).

$$\mathrm{IntegralDomain}(R) \Longrightarrow \forall a,b,c\in R,\ a\neq 0\wedge ab=ac\Rightarrow b=c.$$

Remark (Theorem predicate reading). Nonzero factors can be cancelled in an integral domain.

Remark (Negated quantified statement).

$$\mathrm{IntegralDomain}(R)\wedge\exists a,b,c\in R,\ a\neq 0\wedge ab=ac\wedge b\neq c \Longrightarrow \bot.$$

Remark (Interpretation). Cancellation is recovered in integral domains because nonzero elements are not zero divisors.

Remark (Dependencies).

- Integral domain.
- Zero divisor.

Remark (Proof TODO). Proof exists in the archived chapter and should be migrated to a one-proof-per-file artifact. Source: archive/algebra-and-abstract-structures-legacy.tex.

1.6.9 Fields

Definition — Field

Definition 1.6.15 (Field). A **field** is a set F equipped with two binary operations, addition $+$ and multiplication $\cdot$, such that:

F1. $(F, +)$ is an abelian group with identity 0;

F2. $F \setminus \{0\}$ is an abelian group under multiplication with identity 1;

F3. multiplication distributes over addition.

Remark (Standard quantified statement).

$$\mathrm{Field}(F, +, \cdot, 0, 1) \iff \mathrm{AbelianGroup}(F, +) \wedge \mathrm{AbelianGroup}(F \setminus \{0\}, \cdot) \wedge \mathrm{LeftDistributive}(\cdot, +, F) \wedge \mathrm{RightI}$$

Remark (Definition predicate reading). A field is an additive abelian group whose nonzero elements form a multiplicative abelian group, with multiplication distributing over addition.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Field}(F, +, \cdot, 0, 1) \iff \neg\,\mathrm{AbelianGroup}(F, +) \vee \neg\,\mathrm{AbelianGroup}(F \setminus \{0\}, \cdot) \vee \neg\,\mathrm{LeftDistributive}(\cdot, +, F)$$

Remark (Interpretation). Fields are the algebraic setting where addition, subtraction multiplication, and division by nonzero elements are all available.

Remark (Dependencies).

- Abelian group.
- Distributivity.

Proposition 1.6.16 (Every Field is an Integral Domain). *Every field is an integral domain.*

Remark (Standard quantified statement).

$$\mathrm{Field}(F, +, \cdot, 0, 1) \Longrightarrow \mathrm{IntegralDomain}(F, +, \cdot, 0, 1).$$

Remark (Theorem predicate reading). The field axioms imply the integral-domain axioms.

Remark (Negated quantified statement).

$$\mathrm{Field}(F, +, \cdot, 0, 1) \wedge \neg\,\mathrm{IntegralDomain}(F, +, \cdot, 0, 1) \Longrightarrow \bot.$$

Remark (Interpretation). The existence of inverses for nonzero elements prevents nonzero zero divisors.

Remark (Dependencies).

- Field.
- Integral domain.
- Multiplicative inverse.

Remark (Proof TODO). Proof exists in the archived chapter and should be migrated to a one-proof-per-file artifact. Source: archive/algebra-and-abstract-structures-legacy.tex.

Remark (Examples). The rational, real, and complex numbers are fields. The integers are not a field because nonzero integers such as 2 do not have multiplicativ inverses inside $\mathbb{Z}$.

1.7 Number Systems as Structures

Number Systems as Structures Toolkit

The standard number systems can be read as structures by naming their carrier sets, arithmetic operations, distinguished constants, order, and laws.

1.7.1 The Arithmetic Signature

Definition 1.7.1 (Arithmetic Signature)**.** The **arithmetic signature** is the collection of symbols used to view number systems as structures: arithmetic operations such as $+$ and $\cdot$, distinguished constants such as 0 and 1, and relations such as $\leq$ when order is included.

Remark (Standard quantified statement)*.*

$$\mathcal{L}_{\text{arith}} = \{+, \cdot, 0, 1, \leq, \ldots\},$$

with the visible symbols chosen according to the intended number-system structure.

Remark (Definition predicate reading)*.* The arithmetic signature is the symbol list used before a particular number system interprets those symbols.

Remark (Interpretation)*.* Different number systems interpret the same visible arithmetic symbols on different carrier sets. For example, addition on $\mathbb{N}$ and addition on $\mathbb{R}$ have the same symbol but different domains and different structural consequences.

Remark (Dependencies)*.*

- Structure signature.
- Operations.
- Distinguished constants.

1.7.2 The Natural Numbers as a Structure

$\mathbb{N}$ as a Model of the Arithmetic Signature Toolkit

We exhibit $\mathbb{N}$ as a concrete *model*: fix the arithmetic signature, take the carrier and operations built in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{N}$ obeys the relevant natural-number axioms. The successor fragment recovers the Peano-system view; the additive and multiplicative fragments are obtained by reduction.

Remark (Exposition)*.* A model is a signature together with an interpretation on a carrier. The signature here is the natural-number fragment of the arithmetic signature,

$$\mathcal{L}_{\mathbb{N}} = \{S, +, \cdot, 0, 1\},$$

with unary function symbol S, binary function symbols $+, \cdot$, and constant symbols $0, 1$. The order symbol $<$ is deliberately absent; it enters by expansion below.

Definition 1.7.2 (The Natural Number Model $\mathcal{N}$). The **natural number model** is the $\mathcal{L}_{\mathbb{N}}$-structure

$$\mathcal{N} = \bigl(\mathbb{N};\, S^{\mathcal{N}},\, +^{\mathcal{N}},\, \cdot^{\mathcal{N}},\, 0^{\mathcal{N}},\, 1^{\mathcal{N}}\bigr),$$

whose carrier $|\mathcal{N}| = \mathbb{N}$ is the set of natural numbers constructed in Volume II, and whose symbols are interpreted by the natural-number operations: $S^{\mathcal{N}}$ is successor, $+^{\mathcal{N}}$ is natural-number addition, $\cdot^{\mathcal{N}}$ is natural-number multiplication, and $0^{\mathcal{N}}, 1^{\mathcal{N}}$ are the natural-number constants 0 and 1.

Remark (Standard quantified statement).

$$|\mathcal{N}| = \mathbb{N}, \qquad S^{\mathcal{N}} : \mathbb{N} \to \mathbb{N}, \qquad +^{\mathcal{N}}, \cdot^{\mathcal{N}} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad 0^{\mathcal{N}}, 1^{\mathcal{N}} \in \mathbb{N}.$$

Remark (Interpretation). Nothing here is new mathematics: the carrier, successor, addition, and multiplication are exactly the objects built in Volume II. The model only packages them as a single object $\mathcal{N}$, so that "$\mathbb{N}$" can be handed to any theorem that quantifies over arithmetic structures. The signature is the type; $\mathcal{N}$ is the instance.

Remark (Dependencies).

- Arithmetic Signature.
- Peano System.
- Volume II: construction of $\mathbb{N}$ and the natural-number operations.

Remark (Satisfaction). $\mathcal{N}$ is a *model* of the natural-number axioms:

$$\mathcal{N} \models \Sigma_{\mathbb{N}}, \qquad \text{equivalently} \qquad \mathrm{PeanoSystem}(\mathbb{N}, S, 0).$$

This is not re-proved here. It is exactly the content of $\mathbb{N}$ Is a Peano System in Volume II, which verifies the Peano axioms against the constructed successor operation. The algebraic arithmetic fragment is witnessed separately by $\mathbb{N}$ Is a Commutative Semiring, which verifies the laws of addition and multiplication on $\mathbb{N}$.

Remark (Expansion to the ordered natural-number model). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{N}_{<} = (\mathbb{N}; S, +, \cdot, 0, 1, <), \qquad \mathrm{Expansion}(\mathcal{N}_{<}, \mathcal{N}),$$

in the sense of Structure Expansion: the carrier is unchanged and exactly one new interpreted symbol is added. Here $<^{\mathcal{N}}$ is the usual order on $\mathbb{N}$. That this expansion has the expected order behavior is witnessed by $\mathbb{N}$ Is Well-Ordered in Volume II.

Remark (Reduct to the successor structure). Forgetting $+, \cdot, 1$ leaves the successor reduct

$$\mathcal{N} \restriction \{S, 0\} = (\mathbb{N}; S, 0), \qquad \mathrm{Reduction}\bigl(\mathcal{N} \restriction \{S, 0\},\, \mathcal{N}\bigr),$$

in the sense of Structure Reduction: the same carrier with fewer visible symbols. This reduct is the Peano-system view of the natural numbers.

Remark (Reduct to the additive monoid). Forgetting $S, \cdot, 1$ leaves the additive reduct

$$\mathcal{N} \restriction \{+, 0\} = (\mathbb{N}; +, 0), \qquad \mathrm{Reduction}\bigl(\mathcal{N} \restriction \{+, 0\},\, \mathcal{N}\bigr).$$

This reduct is the additive commutative monoid of natural numbers: it remembers addition and the additive identity, but forgets multiplication and successor as primitive symbols.

Remark (Reduct to the multiplicative monoid). Forgetting $S, +, 0$ leaves the multiplicative reduct

$$\mathcal{N} \restriction \{\cdot, 1\} = (\mathbb{N}; \cdot, 1), \qquad \mathrm{Reduction}\big(\mathcal{N} \restriction \{\cdot, 1\}, \mathcal{N}\big).$$

This reduct is the multiplicative commutative monoid of natural numbers.

Remark (Blueprint position). In the structural blueprint $\mathcal{N}$ is the initial counting structure: it is the first model in the number-system hierarchy and the source of the successor, addition, multiplication, and induction principles used later. It sits at the head of the canonical embedding chain

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The first arrow is the canonical embedding of Volume II, $\mathbb{N}$ Embeds in $\mathbb{Z}$. Passing from $\mathbb{N}$ to $\mathbb{Z}$ supplies additive inverses; passing from $\mathbb{Z}$ to $\mathbb{Q}$ supplies multiplicative inverses for nonzero elements; passing from $\mathbb{Q}$ to $\mathbb{R}$ supplies the analytic completeness needed for real analysis.

Remark (Interpretation). $\mathbb{N}$ is not a group, ring, or field under its usual arithmetic operations: subtraction and division are not closed operations on all of $\mathbb{N}$. Its natural algebraic home is the Peano-system and commutative-semiring layer. This is exactly why the number-system hierarchy proceeds from $\mathbb{N}$ to $\mathbb{Z}$, then to $\mathbb{Q}$, and finally to $\mathbb{R}$.

1.7.3 The Whole Numbers as a Structure

$\mathbb{W}$ as a Zero-Based Counting Model Toolkit

We exhibit the whole numbers as a concrete structure with carrier $\mathbb{W} = \{0, 1, 2, \ldots\}$. In the usual convention this is extensionally the same carrier as $\mathbb{N}$ when $\mathbb{N}$ is zero-based. It is kept here as a named presentation because elementary arithmetic often distinguishes "natural numbers" from "whole numbers" terminologically.

Remark (Exposition). The whole-number signature is the zero-based arithmetic fragment

$$\mathcal{L}_{\mathbb{W}} = \{S, +, \cdot, 0, 1\}.$$

It has unary successor S, binary operations $+, \cdot$, and constants $0, 1$. The order symbol $<$ is absent at first and is added by expansion below.

Definition 1.7.3 (The Whole Number Model $\mathcal{W}$). The **whole number model** is the $\mathcal{L}_{\mathbb{W}}$-structure

$$\mathcal{W} = \big(\mathbb{W};\, S^{\mathcal{W}}, +^{\mathcal{W}}, \cdot^{\mathcal{W}}, 0^{\mathcal{W}}, 1^{\mathcal{W}}\big),$$

where

$$\mathbb{W} = \{0, 1, 2, \ldots\},$$

$S^{\mathcal{W}}$ is successor, $+^{\mathcal{W}}$ is whole-number addition, $\cdot^{\mathcal{W}}$ is whole-number multiplication, and $0^{\mathcal{W}}, 1^{\mathcal{W}}$ are the distinguished whole-number constants.

Remark (Standard quantified statement).

$$|\mathcal{W}| = \mathbb{W}, \qquad S^{\mathcal{W}} : \mathbb{W} \to \mathbb{W}, \qquad +^{\mathcal{W}}, \cdot^{\mathcal{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}, \qquad 0^{\mathcal{W}}, 1^{\mathcal{W}} \in \mathbb{W}.$$

Remark (Interpretation). If the repository defines $\mathbb{N}$ as $\{0, 1, 2, \ldots\}$, then $\mathcal{W}$ is not a new mathematical structure; it is the same zero-based counting structure under the name "whole numbers." If the repository defines $\mathbb{N}$ as $\{1, 2, 3, \ldots\}$, then $\mathbb{W}$ is the zero-adjoined successor structure.

Remark (Satisfaction). The whole-number model satisfies the same zero-based Peano and arithmetic laws as the natural-number model whenever the repository convention identifies

$$\mathbb{W} = \mathbb{N}.$$

Equivalently,

$$\mathcal{W} \models \Sigma_{\mathbb{W}}, \qquad \mathrm{PeanoSystem}(\mathbb{W}, S, 0).$$

This satisfaction claim should be wired to the repository theorem that verifies the Peano-system or commutative-semiring laws for the zero-based counting structure.

Remark (Expansion to the ordered whole-number model). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{W}_{<} = (\mathbb{W}; S, +, \cdot, 0, 1, <), \qquad \mathrm{Expansion}(\mathcal{W}_{<}, \mathcal{W}),$$

where $<^{\mathcal{W}}$ is the usual order on $\{0, 1, 2, \ldots\}$.

Remark (Reduct to the successor structure). Forgetting $+, \cdot, 1$ leaves the successor reduct

$$\mathcal{W} \upharpoonright \{S, 0\} = (\mathbb{W}; S, 0), \qquad \mathrm{Reduction}\big(\mathcal{W} \upharpoonright \{S, 0\},\, \mathcal{W}\big).$$

This is the zero-based counting structure.

Remark (Reduct to the additive monoid). Forgetting $S, \cdot, 1$ leaves

$$\mathcal{W} \upharpoonright \{+, 0\} = (\mathbb{W}; +, 0),$$

the additive commutative monoid of whole numbers.

Remark (Blueprint position). The whole numbers occupy a terminological rather than algebraically new position. They emphasize that zero is included:

$$\mathbb{W} = \{0, 1, 2, \ldots\}.$$

Structurally, they sit at the zero-based counting layer and embed into the integers:

$$\mathbb{W} \hookrightarrow \mathbb{Z}.$$

Passing from $\mathbb{W}$ to $\mathbb{Z}$ supplies additive inverses.

Remark (Interpretation). The whole numbers are closed under addition and multiplicatic contain 0 and 1, and support successor. They are not a group, ring, or field under the usual arithmetic operations because additive inverses and multiplicative inverses are not available internally. Their natural algebraic home is the same commutative-semiring / zero-based Peano layer as the natural-number model.

“‘tex

1.7.4 The Integers as a Structure

$\mathbb{Z}$ as a Model of the Ring Signature Toolkit

We exhibit $\mathbb{Z}$ as a concrete *model*: fix the ring signature, take the carrier and operations constructed in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{Z}$ satisfies the ring axioms. Order is then added by expansion, and various algebraic fragments are recovered by reduction.

Remark (Exposition). A model is a signature together with an interpretation on a carrier. The signature here is the ring fragment of the arithmetic signature,

$$\mathcal{L}_{\text{ring}} = \{+, \cdot, -, 0, 1\},$$

with binary function symbols $+, \cdot$, unary function symbol $-$, and constant symbols $0, 1$.

The order symbol $<$ is deliberately absent. It enters only through expansion.

Definition 1.7.4 (The Integer Model $\mathcal{Z}$). The **integer model** is the $\mathcal{L}_{\text{ring}}$-structure

$$\mathcal{Z} = \left(\mathbb{Z}; +^{\mathcal{Z}}, \cdot^{\mathcal{Z}}, -^{\mathcal{Z}}, 0^{\mathcal{Z}}, 1^{\mathcal{Z}}\right),$$

whose carrier $|\mathcal{Z}| = \mathbb{Z}$ is the set of integers constructed in Volume II, and whose symbols are interpreted by the integer operations.

Thus

$$+^{\mathcal{Z}}$$

is integer addition,

$$\cdot^{\mathcal{Z}}$$

is integer multiplication,

$$-^{\mathcal{Z}}$$

is integer negation, and

$$0^{\mathcal{Z}}, \quad 1^{\mathcal{Z}}$$

are the distinguished integer constants.

Remark (Standard quantified statement).

$$|\mathcal{Z}| = \mathbb{Z},$$

$$+^{\mathcal{Z}}, \cdot^{\mathcal{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},$$

$$-^{\mathcal{Z}} : \mathbb{Z} \to \mathbb{Z},$$

$$0^{\mathcal{Z}}, 1^{\mathcal{Z}} \in \mathbb{Z}.$$

Remark (Interpretation). Nothing new is constructed here.
The carrier and operations were already built in Volume II.
The purpose of the model

$$\mathcal{Z}$$

is to package the integer system as a single mathematical object so that theorems may quantify over rings and then be applied directly to the integers.
The signature is the type.
The structure

$$\mathcal{Z}$$

is the instance.

Remark (Dependencies).
- Arithmetic Signature.
- Ring.
- Volume II construction of $\mathbb{Z}$.
- Integer Operations.

Remark (Satisfaction). The structure

$$\mathcal{Z}$$

is a model of the ring axioms:

$$\mathcal{Z} \models \Sigma_{\text{ring}},$$

equivalently,

$$\text{Ring}(\mathbb{Z}, +, \cdot, 0, 1).$$

This is not re-proved here.
It is exactly the content of
$\mathbb{Z}$ Is a Ring
in Volume II, which verifies each ring axiom against the integer operations.
Likewise,

$$\mathcal{Z} \models \Sigma_{\text{id}},$$

because
$\mathbb{Z}$ Is an Integral Domain
shows that multiplication on the integers has no zero divisors.

Remark (Expansion to the Ordered Ring). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{Z}_{<} = (\mathbb{Z}; +, \cdot, -, 0, 1, <),$$

$$\text{Expansion}(\mathcal{Z}_{<}, \mathcal{Z}),$$

in the sense of Structure Expansion.

The carrier is unchanged and exactly one interpreted symbol is added.
The relation

$$<^{\mathcal{Z}}$$

is the usual integer order determined by the positive cone.
That this expansion satisfies the ordered-ring axioms is witnessed by
$\mathbb{Z}$ Is an Ordered Ring
in Volume II.

Remark (Reduct to the Additive Group). Forgetting

$$\cdot, 1$$

leaves the reduct

$$\mathcal{Z} \restriction \{+, -, 0\} = (\mathbb{Z}; +, -, 0),$$

$$\text{Reduction}\big(\mathcal{Z} \restriction \{+, -, 0\}, \mathcal{Z}\big).$$

This reduct is the additive abelian group of integers.
It remembers addition and additive inverses while ignoring multiplication.

Remark (Reduct to the Multiplicative Monoid). Forgetting

$$+, -, 0$$

leaves

$$\mathcal{Z} \restriction \{\cdot, 1\} = (\mathbb{Z}; \cdot, 1),$$

the multiplicative commutative monoid of integers.

Remark (Blueprint Position). The integer model occupies the next stage after the natural numbers in the number-system hierarchy.
It sits on the embedding chain

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The first embedding is the canonical map

$$n \longmapsto n,$$

witnessed by
$\mathbb{N}$ Embeds in $\mathbb{Z}$.
The second embedding is witnessed by
$\mathbb{Z}$ Embeds in $\mathbb{Q}$.

Remark (Interpretation). The passage

$$\mathbb{N} \longrightarrow \mathbb{Z}$$

supplies exactly one new algebraic capability:

$$\forall a \in \mathbb{Z}, \quad \exists (-a) \in \mathbb{Z}.$$

Every integer has an additive inverse.
This is the structural feature absent from the natural numbers.
Consequently,

$$(\mathbb{N}, +)$$

is only a commutative monoid, whereas

$$(\mathbb{Z}, +)$$

is an abelian group.
The passage

$$\mathbb{Z} \longrightarrow \mathbb{Q}$$

will later supply multiplicative inverses for nonzero elements and thereby upgrade the integer ring into a field.

Remark (Structural Significance). The integer model is the first standard number system that satisfies the full ring signature.

The natural numbers support addition and multiplication, but not additive inverses.

The integers close precisely that gap.

For this reason $\mathbb{Z}$ occupies the ring layer of the structural hierarchy, sitting between the semiring structure of $\mathbb{N}$ and the field structure of $\mathbb{Q}$.

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1.7.5 The Rational Numbers as a Structure

$\mathbb{Q}$ as a Model of the Field Signature Toolkit

We exhibit $\mathbb{Q}$ as a concrete *model*: fix the field signature, take the carrier and operations built in Volume II, interpret each symbol, and discharge satisfaction by citing the Volume II theorem that $\mathbb{Q}$ obeys the field axioms. Order is then added by *expansion*, and the additive group is recovered by *reduction*.

Remark (Exposition). A model is a signature together with an interpretation on a carrier. The signature here is the field fragment of the arithmetic signature,

$$\mathcal{L}_{\text{field}} = \{+, \cdot, -, ^{-1}, 0, 1\},$$

with binary function symbols $+, \cdot$, a unary $-$, a unary $^{-1}$ defined on nonzero elements, and constant symbols $0, 1$. The order symbol $<$ is deliberately absent; it enters by expansion below.

Definition 1.7.5 (The Rational Field Model $\mathcal{Q}$). The **rational field model** is the $\mathcal{L}_{\text{field}}$-structure

$$\mathcal{Q} = \big(\mathbb{Q};\, +^{\mathcal{Q}},\, \cdot^{\mathcal{Q}},\, -^{\mathcal{Q}},\, ^{-1\,\mathcal{Q}},\, 0^{\mathcal{Q}},\, 1^{\mathcal{Q}}\big),$$

whose carrier $|\mathcal{Q}| = \mathbb{Q}$ is the set of rational numbers constructed in Volume II, and whose symbols are interpreted by the rational operations: $+^{\mathcal{Q}}$ and $\cdot^{\mathcal{Q}}$ are rational addition and multiplication, $-^{\mathcal{Q}}$ is rational negation, $^{-1\,\mathcal{Q}}$ is the inverse of a nonzero rational, and $0^{\mathcal{Q}}, 1^{\mathcal{Q}}$ are the rational constants 0 and 1.

Remark (Standard quantified statement).

$$|\mathcal{Q}| = \mathbb{Q}, \qquad +^{\mathcal{Q}}, \cdot^{\mathcal{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}, \qquad -^{\mathcal{Q}} : \mathbb{Q} \to \mathbb{Q}, \qquad 0^{\mathcal{Q}}, 1^{\mathcal{Q}} \in \mathbb{Q}.$$

Remark (Interpretation). Nothing here is new mathematics: the carrier and operations are exactly the ones built in Volume II. The model only packages them as a single object $\mathcal{Q}$, so that "$\mathbb{Q}$" can be handed to any theorem that quantifies over fields, and so the satisfaction statement below has a concrete structure to range over. The signature is the type; $\mathcal{Q}$ is the instance.

Remark (Dependencies).

- Arithmetic Signature.
- Field (Volume II).
- Volume II: construction of $\mathbb{Q}$ and the rational operations.

Remark (Satisfaction). $\mathcal{Q}$ is a *model* of the field axioms:

$$\mathcal{Q} \models \Sigma_{\text{field}}, \qquad \text{equivalently} \qquad \text{Field}(\mathbb{Q}, +, \cdot, 0, 1).$$

This is not re-proved here. It is exactly the content of $\mathbb{Q}$ Is a Field in Volume II, which verifies every field axiom against the rational operations; that theorem is therefore the witness that the interpretation $\mathcal{Q}$ satisfies Σ_{field}. The integral-domain fragment is witnessed separately by $\mathbb{Q}$ Is an Integral Domain.

Remark (Expansion to the ordered field). Adjoining the relation symbol $<$ gives the expansion

$$\mathcal{Q}_< = (\mathbb{Q}; +, \cdot, -, {}^{-1}, 0, 1, <), \qquad \text{Expansion}(\mathcal{Q}_<, \mathcal{Q}),$$

in the sense of Structure Expansion: the carrier is unchanged and exactly one new interpreted symbol is added, where $<^{\mathcal{Q}}$ is the rational order determined by the positive cone. That $\mathcal{Q}_<$ satisfies the ordered-field axioms is witnessed by $\mathbb{Q}$ Is an Ordered Field in Volume II.

Remark (Reduct to the additive group). Forgetting $\cdot, {}^{-1}, 1$ leaves the reduct

$$\mathcal{Q}\restriction\{+, -, 0\} = (\mathbb{Q}; +, -, 0), \qquad \text{Reduction}\big(\mathcal{Q}\restriction\{+, -, 0\},\ \mathcal{Q}\big),$$

in the sense of Structure Reduction: the same carrier with fewer visible symbols. This reduct is the additive abelian group of $\mathbb{Q}$ -- what remains when only additive structure is in view.

Remark (Blueprint position). In the structural blueprint $\mathcal{Q}$ is the *field of fractions* of the integer ring: the smallest field into which the integer model embeds. This places it on the chain

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R},$$

whose arrows are the canonical embeddings of Volume II ($\mathbb{Z}$ Embeds in $\mathbb{Q}$, $\mathbb{Q}$ Embeds in $\mathbb{R}$). Model-theoretically the first embedding presents the integer model as a substructure of the $\{+, \cdot, 0, 1\}$-reduct of $\mathcal{Q}$; passing to the full field $\mathcal{Q}$ supplies the multiplicative inverses the integer model lacks.

Remark (Interpretation). $\mathbb{Q}$ is the first number system whose model satisfies the *full* field signature: $\mathbb{Z}$ is only a ring (no inverses), and $\mathbb{Q}$ closes exactly that gap. Adding $<$ by expansion gives the ordered field; adding analytic completeness -- which is *not* a first-order property of the field signature -- is deferred to the real model in the analysis layer.

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1.7.6 The Real Numbers as a Structure

$\mathbb{R}$ as a Model of the Ordered Field Signature Toolkit

We exhibit $\mathbb{R}$ as a concrete model of the ordered-field signature. The field structure is first-order and is captured by satisfaction of the ordered-field axioms. The completeness of $\mathbb{R}$ is then discussed separately, because it does not belong to the ordinary first-order field signature.

Remark (Exposition). A model consists of a signature together with an interpretation on a carrier.

The signature considered here is the ordered-field signature

$$\mathcal{L}_{\text{of}} = \{+, \cdot, -, ^{-1}, 0, 1, <\}.$$

It contains the field operations together with a distinguished order relation.

Definition 1.7.6 (The Real Number Model $\mathcal{R}$). The **real number model** is the $\mathcal{L}_{\text{of}}$-structure

$$\mathcal{R} = \left(\mathbb{R}; +^{\mathcal{R}}, \cdot^{\mathcal{R}}, -^{\mathcal{R}}, {}^{-1\mathcal{R}}, 0^{\mathcal{R}}, 1^{\mathcal{R}}, <^{\mathcal{R}}\right),$$

whose carrier

$$|\mathcal{R}| = \mathbb{R}$$

is the set of real numbers constructed in Volume II, and whose symbols are interpreted by the real-number operations and order.

Remark (Standard quantified statement).

$$|\mathcal{R}| = \mathbb{R},$$

$$+^{\mathcal{R}}, \cdot^{\mathcal{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R},$$

$$-^{\mathcal{R}} : \mathbb{R} \to \mathbb{R},$$

$${}^{-1\mathcal{R}} : \mathbb{R} \setminus \{0\} \to \mathbb{R},$$

$$0^{\mathcal{R}}, 1^{\mathcal{R}} \in \mathbb{R},$$

$$<^{\mathcal{R}} \subseteq \mathbb{R} \times \mathbb{R}.$$

Remark (Interpretation). The structure

$$\mathcal{R}$$

packages the entire real-number system into a single mathematical object.

The carrier, operations, inverses, and order already exist from the Volume II construction. The model simply presents them as a unified structure that may serve as an argument to general theorems about ordered fields.

Remark (Dependencies).

- Field.
- Positive Cone.
- Volume II construction of $\mathbb{R}$.
- Ordered-field development in Volume II.

Remark (Satisfaction). The structure

$$\mathcal{R}$$

is a model of the ordered-field axioms:

$$\mathcal{R} \models \Sigma_{\mathrm{of}},$$

equivalently,

$$\mathrm{OrderedField}(\mathbb{R}, +, \cdot, <, 0, 1).$$

This is witnessed by
$\mathbb{R}$ Is an Ordered Field
in Volume II.
In particular,

$$\mathcal{R} \models \Sigma_{\mathrm{field}}$$

because
$\mathbb{R}$ Is a Field
verifies the field axioms.

Remark (Completeness Is Separate). The ordered-field structure does *not* yet distinguish $\mathbb{R}$ from every other ordered field.

The defining analytic property of the real numbers is completeness, expressed in Volume II by

$$LeastUpperBoundProperty(\mathbb{R}).$$

Equivalently, every nonempty subset of $\mathbb{R}$ that is bounded above possesses a least upper bound.

This property is witnessed by
Completeness of the Real Numbers.

Remark (Important Structural Distinction). The statement

$$\mathcal{R} \models \Sigma_{\mathrm{of}}$$

belongs to the first-order ordered-field structure.

The completeness property does not belong to the ordinary ordered-field signature.

Consequently the real-number model should be viewed as

$$\texttt{ordered field} + \texttt{completeness}.$$

These are conceptually distinct layers.

Remark (Reduction to the Field Structure). Forgetting the order relation leaves the reduct

$$\mathcal{R} \restriction \{+, \cdot, -, ^{-1}, 0, 1\},$$

which is simply the field structure of the real numbers.

$$\mathrm{Reduction}\Big(\mathcal{R} \restriction \{+, \cdot, -, ^{-1}, 0, 1\}, \mathcal{R}\Big).$$

Remark (Reduction to the Additive Group). Forgetting multiplication and order leaves

$$(\mathbb{R}; +, -, 0),$$

the additive abelian group of real numbers.

Remark (Blueprint Position). The real-number model occupies the final stage of the standard ordered number-system hierarchy:

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}.$$

The final embedding is witnessed by

$\mathbb{Q}$ Embeds in $\mathbb{R}$.

The rationals appear inside $\mathbb{R}$ as a subfield.

Remark (Interpretation). The passage

$$\mathbb{Q} \longrightarrow \mathbb{R}$$

does not add new field operations.

Addition, subtraction, multiplication, and division already exist in $\mathbb{Q}$.

Instead it adds completeness.

This closes the gaps left by the rational numbers and supplies limits, suprema, infima, and the analytic foundation required for calculus and real analysis.

Remark (Structural Significance). The real-number model is simultaneously

- a field,
- an ordered field,
- a complete ordered field.

The first two properties belong to its algebraic structure.

The third is the property that makes real analysis possible.

For this reason $\mathbb{R}$ occupies the analytic layer of the structural hierarchy and serves as the foundational model for limits, continuity, differentiation, and integration.

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1.7.7 The Complex Numbers as a Structure

$\mathbb{C}$ as a Model of the Field Signature Toolkit

We exhibit $\mathbb{C}$ as a concrete field model. The real numbers embed into $\mathbb{C}$, and the resulting structure is the first standard number system that is algebraically closed. Unlike $\mathbb{R}$, however, $\mathbb{C}$ admits no compatible ordered-field expansion.

Remark (Exposition). A model consists of a signature together with an interpretation on a carrier.

The signature considered here is the field signature

$$\mathcal{L}_{\text{field}} = \{+, \cdot, -, ^{-1}, 0, 1\}.$$

It contains the field operations and distinguished constants but no order relation.

Definition 1.7.7 (The Complex Number Model $\mathcal{C}$). The **complex number model** is the $\mathcal{L}_{\text{field}}$-structure

$$\mathcal{C} = \left(\mathbb{C}; +^{\mathcal{C}}, \cdot^{\mathcal{C}}, -^{\mathcal{C}}, {}^{-1\mathcal{C}}, 0^{\mathcal{C}}, 1^{\mathcal{C}}\right),$$

whose carrier

$$|\mathcal{C}| = \mathbb{C}$$

is the set of complex numbers constructed in Volume II, and whose symbols are interpreted by the usual complex-number operations.

Remark (Standard quantified statement).

$$|\mathcal{C}| = \mathbb{C},$$

$$+^{\mathcal{C}}, \cdot^{\mathcal{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C},$$

$$-^{\mathcal{C}} : \mathbb{C} \to \mathbb{C},$$

$${}^{-1\mathcal{C}} : \mathbb{C} \setminus \{0\} \to \mathbb{C},$$

$$0^{\mathcal{C}}, 1^{\mathcal{C}} \in \mathbb{C}.$$

Remark (Interpretation). The structure

$$\mathcal{C}$$

packages the complex-number system into a single mathematical object.

The carrier and operations are those already constructed in Volume II. The model merely records how the field signature is interpreted on that carrier.

Remark (Dependencies).

- Field.
- Volume II construction of $\mathbb{C}$.
- Complex arithmetic.
- The Fundamental Theorem of Algebra.

Remark (Satisfaction). The structure

$$\mathcal{C}$$

is a model of the field axioms:

$$\mathcal{C} \models \Sigma_{\text{field}},$$

equivalently,

$$\mathrm{Field}(\mathbb{C}, +, \cdot, 0, 1).$$

This is witnessed by
$\mathbb{C}$ Is a Field
in Volume II.

Remark (Algebraic Closure). The most important structural property of the complex numbers is not merely that they form a field.

The defining feature of $\mathbb{C}$ is that it is algebraically closed.
Equivalently,

$$\forall p(x) \in \mathbb{C}[x], \qquad \deg p \geq 1 \Longrightarrow \exists z \in \mathbb{C} \quad p(z) = 0.$$

This property is witnessed by
Fundamental Theorem of Algebra.

Remark (Interpretation). Every nonconstant polynomial with complex coefficients has a complex root.

Consequently every polynomial factors completely into linear factors over $\mathbb{C}$.

This is the structural reason the complex numbers occupy the final stage of the standard algebraic number-system hierarchy.

Remark (Real Numbers as a Substructure). The real numbers embed into the complex numbers through the canonical map

$$r \longmapsto r + 0i.$$

This embedding is witnessed by
$\mathbb{R}$ Embeds in $\mathbb{C}$.
Under this embedding the real numbers appear as a subfield of the complex field.

Remark (Reduction to the Additive Group). Forgetting multiplication leaves

$$(\mathbb{C}; +, -, 0),$$

the additive abelian group of complex numbers.

Remark (Reduction to the Multiplicative Group). Restricting to nonzero elements and forgetting addition leaves

$$(\mathbb{C}^\times; \cdot, 1),$$

the multiplicative abelian group of nonzero complex numbers.

Remark (No Ordered-Field Expansion). Unlike $\mathbb{Q}$ and $\mathbb{R}$, the complex field cannot be expanded into an ordered field compatible with its field operations.

Indeed,

$$i^2 = -1.$$

In every ordered field, every square is nonnegative. Hence

$$i^2 \geq 0.$$

But this would imply

$$-1 \geq 0,$$

which contradicts the ordered-field axioms.

Therefore no order relation can be added to the complex field so as to make it an ordered field.

Remark (Blueprint Position). The complex-number model occupies the final stage of the standard number hierarchy:

$$\mathbb{N} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C}.$$

The passage

$$\mathbb{R} \longrightarrow \mathbb{C}$$

supplies solutions to polynomial equations that have no real solutions.

For example,

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$ but has the solutions

$$x = \pm i$$

in $\mathbb{C}$.

Remark (Structural Significance). The complex-number model is

- a field,
- an algebraically closed field,

- not an ordered field.

The real numbers complete the ordered-field hierarchy.

The complex numbers complete the algebraic hierarchy.

For this reason $\mathbb{C}$ serves as the canonical ambient field for algebra, complex analysis, algebraic geometry, and much of modern mathematics.

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1.8 Number Systems as Models

Number Systems as Models Toolkit

Model language separates a signature from its interpretations. Expanding or reducing a structure changes which operations, relations, and constants are visible while keeping the underlying carrier in view.

1.8.1 Each Number System as a Model

Remark (Exposition). Each number system is a model once a signature is fixed. The carrier supplies the elements, and the operations, relations, and constants interpret the symbols of the chosen arithmetic language.

1.8.2 Structure Expansion

Definition 1.8.1 (Structure Expansion). A **structure expansion** is obtained by adding new interpreted symbols to an existing structure while keeping the same carrier set.

Remark (Standard quantified statement).

$$\mathrm{Expansion}(\mathcal{A}', \mathcal{A}) \iff |\mathcal{A}'| = |\mathcal{A}| \text{ and } \mathcal{A}' \text{ interprets all symbols of } \mathcal{A} \text{ plus additional symbo}$$

Remark (Definition predicate reading). $\mathcal{A}'$ expands $\mathcal{A}$ exactly when it keeps the same carrier and adds visible structure.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Expansion}(\mathcal{A}', \mathcal{A})$$

when the carrier changes or the original interpretations are not preserved.

Remark (Interpretation). Adding an order relation to a field signature expands the visible structure: the carrier remains the same, but more predicates or operations are available.

Remark (Dependencies).

- Carrier set.
- Structure signature.

1.8.3 Structure Reduction

Definition 1.8.2 (Structure Reduction). A **structure reduction** is obtained by forgetting some interpreted symbols of a structure while keeping the same carrier set.

Remark (Standard quantified statement).

$$\mathrm{Reduction}(\mathcal{A}_0, \mathcal{A}) \iff |\mathcal{A}_0| = |\mathcal{A}| \text{ and } \mathcal{A}_0 \text{ retains only selected symbols of } \mathcal{A}.$$

Remark (Definition predicate reading). $\mathcal{A}_0$ reduces $\mathcal{A}$ exactly when it keeps the carrier and forgets some visible structure.

Remark (Negated quantified statement).

$$\neg\,\mathrm{Reduction}(\mathcal{A}_0, \mathcal{A})$$

when the carrier changes or the retained interpretations do not agree.

Remark (Interpretation). Forgetting multiplication in a ring leaves its additive group visible. Forgetting order in an ordered field leaves the underlying field structure.

Remark (Dependencies).

- Carrier set.
- Structure signature.

1.8.4 Blueprint Position of Each Number System

Remark (Exposition). The standard number systems occupy different positions in the algebraic blueprint: $\mathbb{Z}$ is a ring, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields, and $\mathbb{C}$ is a field without the usual compatible total order.

Remark (Migration TODO). The archived chapter contains material on the full number-system blueprint table, but it was not migrated in this pass because the archive contains broad hierarchy prose, but a final table should be reconciled with the repository's canonical number-system conventions before migration. Source: archive/algebra-and-abstract-structures-legacy.tex.

Chapter 2

Abstract Algebra

Where You Are in the Journey

Propositional Logic $\rightarrow$ Sets & Functions $\rightarrow$ $\mathbb{Z}$, $\mathbb{R}$ $\rightarrow$ Algebraic Structures $\rightarrow$ Linear Algebra $\rightarrow$ **Abstract Algebra** $\rightarrow$ Algebraic Geometry $\rightarrow$ $\cdots$

How we got here. Algebraic structures introduced groups, rings, and fields via their axioms. Abstract algebra deepens this study: it asks what properties follow from the axioms alone, develops the theory of homomorphisms and isomorphisms, and classifies structures up to structural equivalence.

What this chapter builds. We develop group theory (subgroups, cosets, Lagrange's theorem, normal subgroups, quotient groups, homomorphisms, isomorphisms, and the isomorphism theorems), ring theory (ideals, quotient rings, polynomial rings, and factorisation), and field extensions.

Where this leads. Galois theory uses group theory to answer questions about polynomial equations. Algebraic geometry studies polynomial ideals and their geometric solutions. Number theory uses ring-theoretic tools throughout.

2.1 Abstract Algebra

Structural Roadmap

The development of abstract algebra in this project follows the definition–theorem–structure architecture used throughout the analysis volumes.

The primary driver is *Contemporary Abstract Algebra* by Joseph A. Gallian (7th ed.).

The emphasis is on algebraic structure as an abstraction of symmetry, arithmetic, and linear operations. Each major topic is organized as:

$$\textbf{Definitions} \longrightarrow \textbf{Structural Theorems} \longrightarrow \textbf{Classification and Applications}$$

The global progression follows Gallian in five structural stages:

1. **Foundations: Integers and Equivalence Relations**

(a) Modular arithmetic
(b) Equivalence relations and partitions
(c) Functions and mappings

Structural Theme: Congruence and equivalence encode algebraic structure via partitions. This stage formalizes symmetry at the level of arithmetic.

2. **Groups**

(a) Definition and basic properties
(b) Subgroups
(c) Cyclic groups
(d) Permutation groups
(e) Isomorphisms
(f) Cosets and Lagranges Theorem
(g) Normal subgroups and factor groups
(h) Homomorphisms
(i) Fundamental Theorem of Finite Abelian Groups

Structural Theme: Groups formalize symmetry. Cosets measure deviation from subgroup structure. Normality permits quotient construction. Homomorphisms reveal structural preservation. Classification emerges in the finite abelian case.

3. **Rings**

(a) Definition and examples
(b) Integral domains
(c) Ideals and factor rings
(d) Ring homomorphisms
(e) Polynomial rings
(f) Unique factorization domains
(g) Euclidean domains

Structural Theme: Rings generalize arithmetic. Ideals control structure via quotients. Factorization encodes algebraic rigidity. Euclidean domains permit algorithmic structure.

4. **Fields and Extensions**

(a) Vector spaces
(b) Field extensions
(c) Algebraic extensions
(d) Finite fields

(e) Geometric constructions

Structural Theme: Fields unify arithmetic and linear algebra. Extensions enlarge solvability. Finite fields exhibit deep classification. Constructibility links algebra and geometry.

5. **Special Topics**

   (a) Sylow Theorems
   (b) Finite simple groups
   (c) Generators and relations
   (d) Symmetry groups
   (e) Group actions and counting (Burnside)
   (f) Coding theory
   (g) Galois theory
   (h) Cyclotomic extensions

   Structural Theme: Group actions connect algebra to combinatorics. Sylow theory controls finite structure. Galois theory connects symmetry and solvability. Cyclotomic extensions bridge algebra and number theory.

Remark 2.1.1. Abstract algebra studies algebraic structures defined by operations subject to axioms. Each structure (group, ring, field) is a package consisting of:

$$\texttt{Set} + \texttt{Operations} + \texttt{Axioms}.$$

Remark 2.1.2. The central construction principle is:

$$\textbf{Substructure} \longrightarrow \textbf{Quotient Structure}.$$

Normal subgroups and ideals make quotient constructions possible.

Remark 2.1.3 (Structural Position). Abstract algebra in this project builds on:

- Set theory and logic (Phase 0)
- Linear algebra (vector space structure)
- Real analysis (structural proof discipline)

Group theory forms the conceptual backbone. Ring theory extends arithmetic. Field theory culminates in symmetry of roots (Galois theory).

2.2 Integers

2.2.1 Basic Definitions and Theorems

2.3 Modular Arithmetic

2.3.1 Basic Definitions and Theorems

2.4 Induction

2.4.1 Definitions and Theorems

Proofs

Capstone

Chapter 3

Linear Algebra

Toolkit — Vector Space Preliminaries

Concept family: The concrete foundations for $\mathbf{F}^n$ before the abstract vector space axioms are introduced.

Atomic items in this section:

- Definition: List
- Definition: $\mathbf{F}^n$
- Definition: Addition in $\mathbf{F}^n$
- Theorem: Addition Is Commutative in $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$
- Definition: Additive Inverse in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Key predicate:

$$\text{ListEquality}((x_1,\ldots,x_n),(y_1,\ldots,y_m)) \;:\equiv\; n=m \;\wedge\; \forall k \le n\ (x_k = y_k).$$

Definition (List)

Definition 3.0.1 (List). Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Remark (Standard quantified statement).

$$\text{List}(L) \iff \exists n \in \mathbb{N}\cup\{0\} : \big(\text{Length}(L) = n \wedge L = (x_1, x_2, \ldots, x_n)\big)$$

Remark (Definition predicate reading).

$$\text{List}(L) \iff \exists n \in \mathbb{N}_0, \exists (x_1,\ldots,x_n) : L = (x_1,\ldots,x_n)$$

Remark (Interpretation). A list is the primary mechanism for discussing ordered finite sets of vectors or scalars. Unlike a set, the structural integrity of a list depends on both the identity and the position of its members.

- **Order Sensitivity**: The lists (a, b) and (b, a) are distinct unless $a = b$.
- **Finiteness**: By Axler's convention, a list must have a finite number of elements. Infinite ordered collections are categorized as sequences.
- **Empty List**: A list of length $n = 0$ is called the empty list, denoted by $()$.

Go to proof of equality.

Definition ($\mathbf{F}^n$)

Definition 3.0.2 ($\mathbf{F}^n$). Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Remark (Standard quantified statement).

$$x \in \mathbf{F}^n \iff \exists x_1, \ldots, x_n \in \mathbf{F} \text{ such that } x = (x_1, \ldots, x_n).$$

Remark (Definition predicate reading).

$$\mathrm{Fn}(\mathbf{F}, n) \;:\equiv\; \mathbf{F}^n = \{(x_1, \ldots, x_n) : x_k \in \mathbf{F}\}.$$

Remark (Interpretation). An element of $\mathbf{F}^n$ is an ordered list of n scalars from $\mathbf{F}$. The order matters, and the list is completely specified by its coordinates.

Remark (Dependencies). • Definition: List

Definition (Addition in $\mathbf{F}^n$)

Definition 3.0.3 (Addition in $\mathbf{F}^n$). Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Standard quantified statement).

$$\forall x, y \in \mathbf{F}^n,\quad x = (x_1, \ldots, x_n) \wedge y = (y_1, \ldots, y_n) \Rightarrow x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Interpretation). Addition in $\mathbf{F}^n$ is defined coordinatewise: add the first coordinates, then the second coordinates, and so on through the nth coordinates.

Remark (Dependencies). • Definition: $\mathbf{F}^n$

Theorem (Addition Is Commutative in $\mathbf{F}^n$)

Theorem 3.0.4 (Addition Is Commutative in $\mathbf{F}^n$). *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbf{F}^n,\ x + y = y + x.$$

Remark (Interpretation). Because addition in $\mathbf{F}^n$ is coordinatewise and scalar addition in $\mathbf{F}$ is commutative, vector addition in $\mathbf{F}^n$ is commutative.

Remark (Dependencies).

- Definition: Addition in $\mathbf{F}^n$

Definition (Zero Vector in $\mathbf{F}^n$)

Definition 3.0.5 (Zero Vector in $\mathbf{F}^n$). The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Standard quantified statement).

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Interpretation). The zero vector is the vector whose every coordinate is the scalar 0 in $\mathbf{F}$. It is the additive identity for $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Definition (Additive Inverse in $\mathbf{F}^n$)

Definition 3.0.6 (Additive Inverse in $\mathbf{F}^n$). Let

$$x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \ldots, -x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall x \in \mathbf{F}^n,\\ &\quad x = (x_1, \ldots, x_n) \ \Rightarrow\ -x = (-x_1, \ldots, -x_n).\end{aligned}$$

Remark (Interpretation). The additive inverse in $\mathbf{F}^n$ is formed coordinatewise by taking the additive inverse of each coordinate in $\mathbf{F}$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$

Definition (Scalar Multiplication in $\mathbf{F}^n$)

Definition 3.0.7 (Scalar Multiplication in $\mathbf{F}^n$). Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Remark (Standard quantified statement).

$$\forall \lambda \in \mathbf{F}\ \forall x \in \mathbf{F}^n,$$
$$x = (x_1, \ldots, x_n) \Rightarrow \lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Remark (Interpretation). Scalar multiplication in $\mathbf{F}^n$ is defined coordinatewise: multiply each coordinate of the vector by the scalar λ.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Toolkit — Vector Spaces

Concept family: The abstract vector space axioms and their immediate consequences.

Atomic items in this section:

- Definition: Vector Space (10 axioms)
- Definition: Vector and Point
- Definition: Real Vector Space
- Definition: Complex Vector Space
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$
- Theorem: $a0 = 0$
- Theorem: $(-1)v = -v$

Key predicate:

$$\text{VectorSpace}(V, \mathbf{F}, +, \cdot).$$

The 10 axioms are closure, commutativity, associativity, additive identity, additive inverse, scalar closure, both distributive laws, scalar associativity, and the scalar identity.

Definition (Vector Space)

Definition 3.0.8 (Vector Space). A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Remark (Standard quantified statement).

V is a vector space over $\mathbf{F}$
$\iff$ V is closed under addition and scalar multiplication,
addition on V is commutative and associative,
V has an additive identity and additive inverses,
and scalar multiplication satisfies the two distributive laws,
the associative law for scalar multiplication, and the identity law.

Remark (Definition predicate reading).

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot)$$

Remark (Negated quantified statement).

$\neg\, \mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot) \iff$ at least one of the ten axioms fails for some $u, v, w \in V,\ a, b$

Remark (Interpretation). A vector space is a set whose elements can be added together and multiplied by scalars, with these operations satisfying the natural algebraic rules familiar from $\mathbf{F}^n$.

Remark (Dependencies).
- Definition: $\mathbf{F}^n$ (motivating example)
- Definition: Addition in $\mathbf{F}^n$

- Definition: Scalar Multiplication in $\mathbf{F}^n$

Definition (Vector and Point)

Definition 3.0.9 (Vector and Point). An element of a vector space V is called a **vector**.
When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Remark (Standard quantified statement).

$$v \in V \implies v \text{ is a vector},$$
$$x \in \mathbf{F}^n \implies x \text{ is a point of } \mathbf{F}^n.$$

Remark (Interpretation). The word *vector* emphasizes the algebraic role of an element of V. In $\mathbf{F}^n$, the same object can also be viewed geometrically as a point with coordinates in $\mathbf{F}$.

Remark (Dependencies).
- Definition: Vector Space

Definition (Real Vector Space)

Definition 3.0.10 (Real Vector Space). A **real vector space** is a vector space over $\mathbb{R}$.

Remark (Standard quantified statement).

$$V \text{ is a real vector space} \iff V \text{ is a vector space over } \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealVectorSpace}(V) :\equiv \mathrm{VectorSpace}(V, \mathbb{R}, +, \cdot).$$

Remark (Interpretation). In a real vector space, the scalars used in scalar multiplication are real numbers.

Remark (Dependencies).
- Definition: Vector Space

Definition (Complex Vector Space)

Definition 3.0.11 (Complex Vector Space). A **complex vector space** is a vector space over $\mathbb{C}$.

Remark (Standard quantified statement).

$$V \text{ is a complex vector space} \iff V \text{ is a vector space over } \mathbb{C}.$$

Remark (Definition predicate reading).

$$\mathrm{ComplexVectorSpace}(V) :\equiv \mathrm{VectorSpace}(V, \mathbb{C}, +, \cdot).$$

Remark (Interpretation). In a complex vector space, the scalars used in scalar multiplication are complex numbers.

Remark (Dependencies).
- Definition: Vector Space

Remark (Notation). For each positive integer m, the notation $\mathbf{F}^m$ denotes the vector space of all lists of length m of elements of $\mathbf{F}$.

The notation $\mathbf{F}^\infty$ denotes the set of all infinite lists

$$(x_1, x_2, x_3, \dots)$$

with each $x_j \in \mathbf{F}$, equipped with coordinatewise addition and coordinatewise scalar multiplication.

Remark (Coordinates in Vector Spaces). A pointwise proof is available when vectors are given explicitly by coordinates, as in $\mathbf{F}^n$, because the operations are defined coordinatewise. In an arbitrary vector space, vectors need not come with coordinates, so proofs must use the vector space axioms rather than coordinate calculations.

Theorem (Unique Additive Identity)

Theorem 3.0.12 (Unique Additive Identity). *A vector space has a unique additive identity. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{If } 0, 0' \in V \text{ each satisfy} \\
&\qquad \forall v \in V,\ v + 0 = v \quad \text{and} \quad \forall v \in V,\ v + 0' = v, \\
&\text{then } 0 = 0'.
\end{aligned}$$

Remark (Interpretation). Although the vector space axioms assert the existence of an additive identity, they force that identity to be unique.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Unique Additive Inverse)

Theorem 3.0.13 (Unique Additive Inverse). *Every vector in a vector space has a unique additive inverse. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall v \in V,\ \text{if } w, u \in V \text{ satisfy } v + w = 0 \text{ and } v + u = 0, \\
&\quad \text{then } w = u.
\end{aligned}$$

Remark (Interpretation). The vector space axioms guarantee that each vector has an additive inverse, and that inverse is determined uniquely by the requirement that the sum be 0.

Remark (Notation). If $v \in V$, then the unique additive inverse of v is denoted by $-v$.

If $w, v \in V$, then

$$w - v \ := \ w + (-v).$$

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Identity

Theorem $(0v = 0)$

Theorem 3.0.14 $(0v = 0)$. *For every* $v \in V$*,*

$$0v = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ 0v = 0.$$

Remark (Interpretation). Multiplying any vector by the scalar 0 gives the zero vector. The 0 on the left is the scalar zero in $\mathbf{F}$; the 0 on the right is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem $(a0 = 0)$

Theorem 3.0.15 $(a0 = 0)$. *For every* $a \in \mathbf{F}$*,*

$$a0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbf{F},\ a0 = 0.$$

Remark (Interpretation). Every scalar sends the zero vector to the zero vector. The 0 on the right of a is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem $((-1)v = -v)$

Theorem 3.0.16 $((-1)v = -v)$. *For every* $v \in V$*,*

$$(-1)v = -v.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ (-1)v = -v.$$

Remark (Interpretation). Scalar multiplication by -1 produces the additive inverse of a vector. This justifies the notation $-v$ for the additive inverse: it is literally (-1) times v.

Remark (Dependencies).

- Definition: Vector Space

- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

Toolkit — Vector Subspaces

Concept family: Subspaces, their sums, and direct sums.

Atomic items in this section:

- Definition: Subspace
- Theorem: Conditions for a Subspace
- Definition: Sum of Subspaces
- Theorem: Sum of Subspaces Is the Smallest Containing Subspace
- Definition: Direct Sum
- Theorem: Condition for a Direct Sum
- Theorem: Direct Sum of Two Subspaces

Key predicates:

$$\mathrm{Subspace}(U,V),\quad \mathrm{ClosedUnderAddition}(U),\quad \mathrm{ClosedUnderScalarMultiplication}(U,\mathbf{F}),\quad \mathrm{Dire}$$

Key test: U is a subspace $\iff$ $0 \in U$, U closed under $+$, U closed under scalar multiplication.

Definition (Subspace)

Definition 3.0.17 (Subspace). Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad U \subseteq V \text{ and } U \text{ is a vector space over } \mathbf{F} \\ &\quad \text{under the operations inherited from } V. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Subspace}(U,V) \;:\equiv\; U \subseteq V \;\wedge\; \mathrm{VectorSpace}(U,\mathbf{F},+,\cdot).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Subspace}(U,V) \iff U \not\subseteq V \;\vee\; \neg\,\mathrm{VectorSpace}(U,\mathbf{F},+,\cdot).$$

Remark (Interpretation). A subspace is a subset of a vector space that is closed under the vector-space operations and therefore forms a vector space in its own right.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Conditions for a Subspace)

Theorem 3.0.18 (Conditions for a Subspace). *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$;

(ii) for all $u, v \in U$, *one has* $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ *and all* $u \in U$, *one has* $au \in U$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad 0 \in U \\ &\quad \wedge \ \text{ClosedUnderAddition}(U) \\ &\quad \wedge \ \text{ClosedUnderScalarMultiplication}(U, \mathbf{F}). \end{aligned}$$

Remark (Contrapositive quantified statement).

$$\neg \text{Subspace}(U, V) \iff 0 \notin U \vee \neg \text{ClosedUnderAddition}(U) \vee \neg \text{ClosedUnderScalarMultiplication}(U,$$

Remark (Failure modes). U fails to be a subspace via exactly one of three routes: it omits the zero vector, it contains two vectors whose sum falls outside U, or it contains a vector and a scalar whose product falls outside U.

Remark (Interpretation). To check that a subset is a subspace, it suffices to verify the three conditions. The remaining seven vector space axioms are inherited automatically from V since U uses the same operations. The three conditions are minimal: condition (i) ensures U is non-empty; (ii) and (iii) ensure the inherited operations stay within U.

Remark (Dependencies).

- Definition: Subspace
- Definition: Vector Space

Definition (Sum of Subspaces)

Definition 3.0.19 (Sum of Subspaces). Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} &v \in U_1 + \cdots + U_m \\ &\quad \iff \quad \exists u_1 \in U_1 \ \cdots \ \exists u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{SumOfSubspaces}(W, \{U_1, \dots, U_m\}, V) \;:\equiv\; W = U_1 + \cdots + U_m.$$

Remark (Interpretation). The sum of subspaces consists of all vectors that can be built by adding together one vector from each subspace. Vectors from different subspaces interact; the representation $v = u_1 + \cdots + u_m$ need not be unique.

Remark (Dependencies).

- Definition: Subspace

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)

Theorem 3.0.20 (Sum of Subspaces Is the Smallest Containing Subspace). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \dots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \dots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Subspace}(U_1 + \cdots + U_m,\, V),\\
&U_j \subseteq U_1 + \cdots + U_m \quad \text{for } j = 1, \dots, m,\\
&\forall W \subseteq V \,\Big[\big(W \text{ is a subspace of } V \wedge U_1 \subseteq W \wedge \cdots \wedge U_m \subseteq W\big)\\
&\qquad\qquad \Rightarrow\; U_1 + \cdots + U_m \subseteq W\Big].
\end{aligned}$$

Remark (Interpretation). The sum is the smallest subspace containing all the given subspaces: it is their join in the lattice of subspaces of V. Any subspace containing all of $U_1, \dots, U_m$ must contain every sum $u_1 + \cdots + u_m$, so it contains the sum subspace.

Remark (Dependencies).

- Definition: Sum of Subspaces
- Theorem: Conditions for a Subspace

Definition (Direct Sum)

Definition 3.0.21 (Direct Sum). Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \dots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \dots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Remark (Standard quantified statement).

$$\begin{aligned} V &= U_1 \oplus \cdots \oplus U_m \\ &\iff \quad V = U_1 + \cdots + U_m \\ &\qquad \wedge\ \forall v \in V\ \exists!\, u_1 \in U_1\ \cdots\ \exists!\, u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \ :\equiv\ V = U_1 \oplus \cdots \oplus U_m.$$

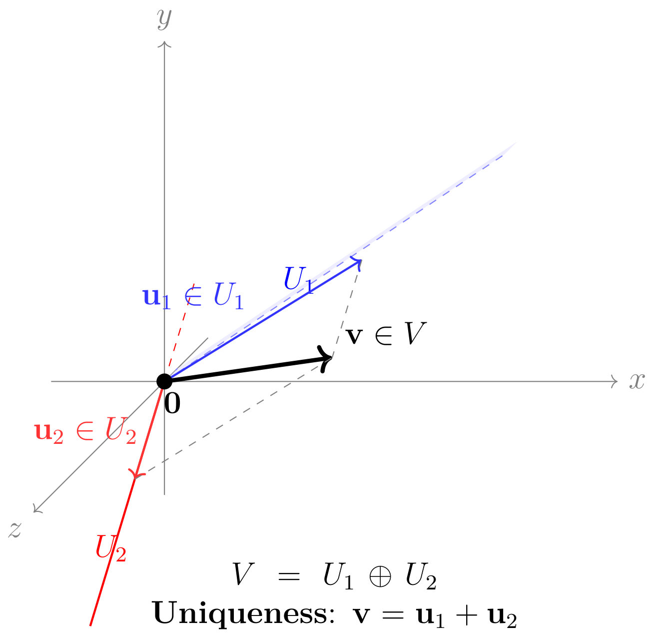
Remark (Negated quantified statement).

$$\neg\, \mathrm{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \iff \exists v \in V\ \exists \text{ two distinct decompositions of } v.$$

Remark (Interpretation). A direct sum is a sum in which every vector has a unique decomposition into pieces coming from the given subspaces. The uniqueness condition is the structural rigidity that makes direct sums so useful it means the subspaces contribute independently, with no redundancy.

Remark (Dependencies).

- Definition: Sum of Subspaces



$V \ = \ U_1 \oplus U_2$
Uniqueness: $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2$
Trivial Intersection:
$U_1 \cap U_2 \ = \ \{\mathbf{0}\}$

Theorem (Condition for a Direct Sum)

Theorem 3.0.22 (Condition for a Direct Sum). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \ldots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &V = U_1 \oplus \cdots \oplus U_m \\ &\quad \iff \quad \forall u_1 \in U_1 \ \cdots \ \forall u_m \in U_m \\ &\qquad \big(u_1 + \cdots + u_m = 0 \Rightarrow u_1 = \cdots = u_m = 0\big). \end{aligned}$$

Remark (Interpretation). A sum is direct exactly when the zero vector has only the trivial decomposition as a sum of vectors from the given subspaces. This is the practical test: rather than verifying uniqueness for every $v \in V$, it suffices to check uniqueness at $v = 0$.

Remark (Dependencies).

- Definition: Direct Sum
- Theorem: Unique Additive Identity

Theorem (Direct Sum of Two Subspaces)

Theorem 3.0.23 (Direct Sum of Two Subspaces). *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U + W = U \oplus W \\ &\quad \iff \quad U \cap W = \{0\}. \end{aligned}$$

Remark (Contrapositive quantified statement).

$$U \cap W \neq \{0\} \iff U + W \neq U \oplus W.$$

If the two subspaces share a nonzero vector, the sum is not direct.

Remark (Interpretation). For two subspaces, directness is equivalent to saying they overlap only in the zero vector. This gives a simple geometric test: draw both subspaces and check whether their intersection is just the origin. The result does not generalise to three or more subspaces -- it is possible for $U_1 \cap U_2$, $U_1 \cap U_3$, and $U_2 \cap U_3$ to all equal $\{0\}$ yet $U_1 + U_2 + U_3$ still fail to be a direct sum.

Remark (Dependencies).

- Theorem: Condition for a Direct Sum
- Definition: Direct Sum

Proofs

Remark (Return). Return to Definition of List

Theorem (Equality of Lists). *Two lists $L = (x_1, \ldots, x_n)$ and $M = (y_1, \ldots, y_m)$ are equal if and only if $n = m$ and $x_j = y_j$ for all $j \in \{1, \ldots, n\}$.*

Proof. **Professional Standard Proof.**
A list of length n is formally defined as a function $f : \{1, \ldots, n\} \to S$. By the axiom of extensionality for functions, two functions f and g are equal if and only if they have the same domain and $f(j) = g(j)$ for all j in that domain. Let L be the function $j \mapsto x_j$ and M be $j \mapsto y_j$. Then $L = M$ if and only if their domains $\{1, \ldots, n\}$ and $\{1, \ldots, m\}$ are equal (implying $n = m$) and $x_j = y_j$ for all $j \in \{1, \ldots, n\}$. □

Proof. **Detailed Learning Proof.**

Step 1. Address the global constraint (Length). Before examining individual elements, we observe that the length n is an intrinsic property of the list. If $n \neq m$, the lists are distinct objects in different spaces (e.g., $\mathbb{F}^n$ vs $\mathbb{F}^m$). Thus, $n = m$ is a necessary condition.

Step 2. Address the local constraint (Positional Identity). Since lists are ordered, identity is not just about having the same "pool" of elements (like a set), but about the mapping of indices to values. We define equality element-wise:

$$\forall j \in \{1, \ldots, n\}, x_j = y_j$$

If there exists even one index k such that $x_k \neq y_k$, the mapping is different, and thus the lists are not equal.

Step 3. Critique of the Induction/Summation approach. While one could define a "difference sum" $D = \sum_{j=1}^{n} \delta(x_j, y_j)$ where δ returns 1 if elements differ, this assumes the elements exist in a space where addition is defined. The functional definition used in Step 2 is more robust because it applies even if the list contains non-mathematical objects. □

Remark (Proof structure). The proof relies on the definition of a list as a finite sequence (function) and uses the principle of functional extensionality.

Remark (Dependencies).

- Definition of List
- FunctionEquality (Volume I)

Remark (Return). Return to Theorem

Theorem (Addition Is Commutative in $\mathbf{F}^n$). *For all* $x, y \in \mathbf{F}^n$,

$$x + y = y + x.$$

Proof. Let $x, y \in \mathbf{F}^n$. Write

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n),$$

where $x_1, \ldots, x_n \in \mathbf{F}$ and $y_1, \ldots, y_n \in \mathbf{F}$.

By the definition of addition in $\mathbf{F}^n$,

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Because addition in $\mathbf{F}$ is commutative, for each $j = 1, \ldots, n$ we have

$$x_j + y_j = y_j + x_j.$$

Hence

$$(x_1 + y_1, \ldots, x_n + y_n) = (y_1 + x_1, \ldots, y_n + x_n).$$

Applying the definition of addition in $\mathbf{F}^n$ again gives

$$(y_1 + x_1, \ldots, y_n + x_n) = y + x.$$

Therefore

$$x + y = y + x.$$

$\square$

Remark (Return). Return to Theorem

Theorem (Unique Additive Identity). *If* $0, 0' \in V$ *each satisfy*

$$\forall v \in V, \quad v + 0 = v \qquad \text{and} \qquad \forall v \in V, \quad v + 0' = v,$$

then

$$0 = 0'.$$

Proof. **Professional Standard Proof.**
Because 0 is an additive identity,

$$0 + 0' = 0'.$$

Because $0'$ is an additive identity,

$$0 + 0' = 0.$$

Hence

$$0 = 0'.$$

Therefore the additive identity is unique. □

Proof. **Detailed Learning Proof.**

Step 1. Start from the assumption that both 0 and $0'$ act as additive identities.
This means that for every $v \in V$,

$$v + 0 = v \qquad \text{and} \qquad v + 0' = v.$$

Step 2. Apply the first identity law to the vector $0'$.
Since 0 is an additive identity,

$$0' + 0 = 0'.$$

By commutativity of addition,

$$0 + 0' = 0'.$$

Step 3. Apply the second identity law to the vector 0.
Since $0'$ is an additive identity,

$$0 + 0' = 0.$$

Step 4. Compare the two values of the same sum.
The expression $0 + 0'$ equals both $0'$ and 0. Therefore

$$0 = 0'.$$

Step 5. State the conclusion.
There cannot be two different additive identities. Hence the additive identity is unique. □

Remark (Proof structure). The argument is direct. The same expression $0 + 0'$ is evaluated using each identity law, and the two resulting values are equated.

Remark (Dependencies).

- Definition of Vector Space

Remark (Return). Return to Theorem

Theorem (Unique Additive Inverse). *For every $v \in V$, if $v_0, v_1 \in V$ satisfy*

$$v + v_0 = 0 \qquad \textit{and} \qquad v + v_1 = 0,$$

then

$$v_0 = v_1.$$

Proof. **Professional Standard Proof.**
Let $v \in V$, and let $v_0, v_1 \in V$ satisfy

$$v + v_0 = 0 \qquad \text{and} \qquad v + v_1 = 0.$$

Add v_1 to both sides of $v + v_0 = 0$:

$$v_1 + (v + v_0) = v_1 + 0.$$

By associativity,

$$(v_1 + v) + v_0 = v_1.$$

By commutativity,

$$v_1 + v = v + v_1 = 0.$$

Hence

$$0 + v_0 = v_1,$$

so

$$v_0 = v_1.$$

Therefore the additive inverse of v is unique. □

Proof. **Detailed Learning Proof.**

Step 1. Fix a vector and two candidate inverses.
Let $v \in V$, and suppose $v_0, v_1 \in V$ satisfy

$$v + v_0 = 0 \qquad \text{and} \qquad v + v_1 = 0.$$

The goal is to show that these two candidates are equal.

Step 2. Start from one inverse equation and add the other inverse to both sides.
From

$$v + v_0 = 0,$$

add v_1 to both sides:

$$v_1 + (v + v_0) = v_1 + 0.$$

Step 3. Simplify the right-hand side.
Because 0 is the additive identity,

$$v_1 + 0 = v_1.$$

So

$$v_1 + (v + v_0) = v_1.$$

Step 4. Reassociate the left-hand side.
By associativity of addition,

$$(v_1 + v) + v_0 = v_1.$$

Step 5. Use commutativity to bring the known inverse relation into view.
By commutativity,

$$v_1 + v = v + v_1.$$

But v_1 is an additive inverse of v, so

$$v + v_1 = 0.$$

Hence

$$(v_1 + v) + v_0 = 0 + v_0.$$

Therefore

$$0 + v_0 = v_1.$$

Step 6. Finish with the identity law.
Because 0 is the additive identity,

$$0 + v_0 = v_0.$$

So

$$v_0 = v_1.$$

Step 7. State the conclusion.
Any two additive inverses of the same vector must agree. Therefore every vector has a unique additive inverse. □

Remark (Proof structure). The proof is direct. One inverse equation is modified by adding the other inverse, and the vector space axioms reduce the resulting expression until the two candidates are shown to be equal.

Remark (Dependencies).

- Definition of Vector Space
- Unique Additive Identity

Remark (Return). Return to Theorem

Theorem ($0v = 0$). *For every* $v \in V$*,*

$$0v = 0.$$

Proof. **Professional Standard Proof.**
Let $v \in V$. By distributivity,

$$(0 + 0)v = 0v + 0v.$$

Because $0 + 0 = 0$ in $\mathbf{F}$, this becomes

$$0v = 0v + 0v.$$

Add the additive inverse of $0v$ to both sides:

$$-(0v) + 0v = -(0v) + (0v + 0v).$$

By associativity,

$$0 = \big(-(0v) + 0v\big) + 0v = 0 + 0v = 0v.$$

Therefore

$$0v = 0.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Fix an arbitrary vector.
Let $v \in V$. The goal is to determine the vector $0v$.
Step 2. Use distributivity over scalar addition.
The vector space axiom gives

$$(a + b)v = av + bv$$

for all $a, b \in \mathbf{F}$. Apply this with $a = 0$ and $b = 0$:

$$(0 + 0)v = 0v + 0v.$$

Step 3. Simplify the scalar expression.
In the field $\mathbf{F}$,

$$0 + 0 = 0.$$

Hence the previous equation becomes

$$0v = 0v + 0v.$$

Step 4. Isolate $0v$ by adding its additive inverse to both sides.
Because $0v \in V$, it has an additive inverse $-(0v)$. Add $-(0v)$ to both sides:

$$-(0v) + 0v = -(0v) + (0v + 0v).$$

Step 5. Simplify both sides.
On the left,

$$-(0v) + 0v = 0.$$

On the right, associativity gives

$$-(0v) + (0v + 0v) = \big(-(0v) + 0v\big) + 0v = 0 + 0v = 0v.$$

Therefore

$$0 = 0v.$$

Step 6. State the conclusion.
Thus scalar multiplication by the scalar zero yields the zero vector:

$$0v = 0.$$

□

Remark (Proof structure). The proof is direct. Distributivity first produces an equation of the form $w = w+w$ with $w = 0v$, and the additive inverse of w is then used to force $w = 0$.

Remark (Dependencies).

- Definition of Vector Space
- Unique Additive Inverse

Remark (Return). Return to Theorem

Theorem ($a0 = 0$). *For every* $a \in \mathbf{F}$*,*

$$a0 = 0.$$

Proof. **Professional Standard Proof.**
Let $a \in \mathbf{F}$. By distributivity,

$$a(0 + 0) = a0 + a0.$$

Because $0 + 0 = 0$ in V, this becomes

$$a0 = a0 + a0.$$

Add the additive inverse of $a0$ to both sides:

$$-(a0) + a0 = -(a0) + (a0 + a0).$$

By associativity,

$$0 = \big(-(a0) + a0\big) + a0 = 0 + a0 = a0.$$

Therefore

$$a0 = 0.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Fix an arbitrary scalar.
Let $a \in \mathbf{F}$. The goal is to determine the vector $a0$.
Step 2. Use distributivity over vector addition.
The vector space axiom gives

$$a(u + v) = au + av$$

for all $u, v \in V$. Apply this with $u = 0$ and $v = 0$:

$$a(0 + 0) = a0 + a0.$$

Step 3. Simplify the vector expression inside the parentheses.
Because 0 is the additive identity in V,

$$0 + 0 = 0.$$

Hence

$$a0 = a0 + a0.$$

Step 4. Add the additive inverse of $a0$ to both sides.
Because $a0 \in V$, it has an additive inverse $-(a0)$. Adding this vector to both sides gives

$$-(a0) + a0 = -(a0) + (a0 + a0).$$

Step 5. Simplify both sides.
On the left,

$$-(a0) + a0 = 0.$$

On the right, associativity gives

$$-(a0) + (a0 + a0) = \big(-(a0) + a0\big) + a0 = 0 + a0 = a0.$$

Therefore

$$0 = a0.$$

Step 6. State the conclusion.
Thus every scalar sends the zero vector to the zero vector:

$$a0 = 0.$$

□

Remark (Proof structure). The proof is direct. Distributivity yields an equation of the form $w = w + w$ with $w = a0$, and additive-inverse cancellation then forces $w = 0$.

Remark (Dependencies).

- Definition of Vector Space
- Unique Additive Inverse

Remark (Return). Return to Theorem

Theorem ($(-1)v = -v$). *For every* $v \in V$,

$$(-1)v = -v.$$

Proof. **Professional Standard Proof.**
Let $v \in V$. Then

$$v + (-1)v = 1v + (-1)v.$$

By distributivity,

$$1v + (-1)v = (1 + (-1))v = 0v.$$

By $0v = 0$, it follows that

$$v + (-1)v = 0.$$

Thus $(-1)v$ is an additive inverse of v. By uniqueness of additive inverses,

$$(-1)v = -v.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Fix an arbitrary vector.
Let $v \in V$. The goal is to show that scalar multiplication by -1 produces the additive inverse of v.

Step 2. Rewrite v as $1v$.
By the scalar identity axiom,

$$1v = v.$$

Therefore

$$v + (-1)v = 1v + (-1)v.$$

Step 3. Use distributivity over scalar addition.
Applying distributivity gives

$$1v + (-1)v = (1 + (-1))v.$$

Because $1 + (-1) = 0$ in $\mathbf{F}$, this becomes

$$1v + (-1)v = 0v.$$

Step 4. Use the theorem $0v = 0$.
By $0v = 0$,

$$0v = 0.$$

Hence

$$v + (-1)v = 0.$$

Step 5. Interpret the resulting equation.
The equation

$$v + (-1)v = 0$$

shows that $(-1)v$ is an additive inverse of v.

Step 6. Use uniqueness of additive inverses.
The vector $-v$ is defined to be the unique additive inverse of v. Since $(-1)v$ is also an additive inverse of v, uniqueness implies

$$(-1)v = -v.$$

Step 7. State the conclusion.
Thus scalar multiplication by -1 yields the additive inverse of a vector. □

Remark (Proof structure). The proof is direct. It shows first that $(-1)v$ satisfies the defining property of an additive inverse, and then invokes uniqueness of additive inverses.

Remark (Dependencies).

- Definition of Vector Space
- $0v = 0$
- Unique Additive Inverse

Remark (Return). Return to Theorem

Theorem (Conditions for a Subspace). *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if: (i) $0 \in U$; (ii) $\forall u, v \in U\ (u + v \in U)$; (iii) $\forall a \in \mathbf{F}\ \forall u \in U\ (au \in U)$.*

Proof. TODO ∎

Remark (Return). Return to Theorem

Theorem (Sum of Subspaces Is the Smallest Containing Subspace). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each U_j, and is contained in every subspace of V that contains all of $U_1, \ldots, U_m$.*

Proof. TODO $\square$

Remark (Return). Return to Theorem

Theorem (Condition for a Direct Sum). *Let $V = U_1 + \cdots + U_m$ where each U_j is a subspace of V. Then $V = U_1 \oplus \cdots \oplus U_m$ if and only if $u_1 + \cdots + u_m = 0$ with $u_j \in U_j$ implies $u_1 = \cdots = u_m = 0$.*

Proof. TODO ∎

Remark (Return). Return to Theorem

Theorem (Direct Sum of Two Subspaces). *Let U and W be subspaces of a vector space V. Then $U + W = U \oplus W$ if and only if $U \cap W = \{0\}$.*

Proof. TODO $\square$

Capstone

Remark (Return). Return to Chapter: Linear Algebra

Capstone -- Linear Algebra

Remark (Capstone Problem). Let V be a vector space over $\mathbf{F}$, and throughout let U, U_1, U_2, W denote subspaces of V.

(a) **(Subspace criterion from the three conditions.)** Prove directly from the definition of a vector space that if a non-empty subset $U \subseteq V$ satisfies conditions (ii) and (iii) of the Subspace Conditions Theorem -- that is, U is closed under addition and scalar multiplication -- then $0 \in U$ automatically. Deduce that for non-empty subsets, condition (i) is redundant and can be dropped.

Hint: If $u \in U$, consider $0 \cdot u$.

(b) **(Direct sum via the zero-decomposition test.)** Let $V = U_1 + U_2$.

(i) Use the Condition for a Direct Sum to prove that $V = U_1 \oplus U_2$ if and only if $U_1 \cap U_2 = \{0\}$.

(ii) Exhibit subspaces U_1, U_2, U_3 of $\mathbf{F}^2$ such that $U_i \cap U_j = \{0\}$ for all $i \neq j$, yet $U_1 + U_2 + U_3 \neq U_1 \oplus U_2 \oplus U_3$. Explain which hypothesis of the Condition for a Direct Sum fails.

(c) **(The zero vector is unique and absorbing.)** Using only the vector space axioms and the theorems proved in this chapter, prove the following chain:

(i) $0v = 0$ for every $v \in V$ (the zero *scalar* kills every vector).

(ii) $a0 = 0$ for every $a \in \mathbf{F}$ (every scalar kills the zero *vector*).

(iii) $(-1)v = -v$ for every $v \in V$.

(iv) Conclude: $(-1)(-1)v = v$. State which axiom and which theorem from (iii) you use at each step.

(d) **(Subspaces and the subspace lattice.)**

(i) Prove that the intersection $U_1 \cap U_2$ of any two subspaces of V is itself a subspace of V.

(ii) Prove that the union $U_1 \cup U_2$ is a subspace of V if and only if $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$.

(iii) Use part (i) to show that the set of all subspaces of V, ordered by inclusion, forms a lattice with meet $U_1 \wedge U_2 = U_1 \cap U_2$ and join $U_1 \vee U_2 = U_1 + U_2$.

(e) **(Connection to Brownian motion on the torus.)** The state space of n-particle Brownian motion on the torus $\mathbb{T}^2$ is $(\mathbb{T}^2)^n \cong (\mathbb{R}^2/\mathbb{Z}^2)^n$. The tangent space at any point is $(\mathbb{R}^2)^n$, a real vector space of dimension $2n$.

(i) Identify the subspace $D \subseteq (\mathbb{R}^2)^n$ of displacement vectors of the form $(v, v, \ldots, v)$ with $v \in \mathbb{R}^2$ (all particles moving identically). What is $\dim D$?

(ii) Identify the subspace $R \subseteq (\mathbb{R}^2)^n$ of relative displacement vectors of the form $(v_1, \dots, v_n)$ with $v_1 + \cdots + v_n = 0$. What is $\dim R$?

(iii) Show that $(\mathbb{R}^2)^n = D \oplus R$ by verifying the direct sum conditions. This decomposition separates the center-of-mass motion (governed by D) from the relative (chase) dynamics (governed by R) in the delay-pursuit simulation.

Concepts synthesized: Subspace criterion (three conditions), Conditions for a Direct Sum, Direct Sum of Two Subspaces ($U \cap W = \{0\}$), Unique Additive Identity, Unique Additive Inverse, $0v = 0$, $a0 = 0$, $(-1)v = -v$, Sum of Subspaces Is the Smallest Containing Subspace.

Connection to simulation. Part (e) is the exact decomposition used to structure the Brownian motion simulation. The center-of-mass component D drifts freely and contributes only to the overall torus translation; the relative component R governs the inter-particle distances and is where the delay-pursuit dynamics live. Projecting the SDE onto D and R separately possible precisely because $(\mathbb{R}^2)^n = D \oplus R$ is what allows the GPU shader to decouple the simulation into an independent drift pass and a relative-position update pass.

Proof. **Solution.**

Part (a): Condition (i) is redundant for non-empty subsets.

Suppose $U \neq \emptyset$ and U is closed under addition and scalar multiplication. Pick any $u \in U$ (possible since U is non-empty). By closure under scalar multiplication with scalar $0 \in \mathbf{F}$: $0 \cdot u \in U$. By the theorem $0v = 0$ (proved from the vector space axioms), $0 \cdot u = 0$. Therefore $0 \in U$.

Hence for non-empty subsets, conditions (ii) and (iii) imply (i). The three-condition criterion can be streamlined to: $U \neq \emptyset$, closed under addition, closed under scalar multiplication.

Part (b): Direct sum and the intersection test.

(i) This is the Direct Sum of Two Subspaces theorem. The key direction is the Condition for a Direct Sum: $V = U_1 \oplus U_2$ iff whenever $u_1 + u_2 = 0$ with $u_i \in U_i$, one has $u_1 = u_2 = 0$. If $v \in U_1 \cap U_2$, then $v + (-v) = 0$ with $v \in U_1$ and $-v \in U_2$ (since U_2 is a subspace). By the zero-decomposition test, $v = 0$. Conversely, if $U_1 \cap U_2 = \{0\}$ and $u_1 + u_2 = 0$, then $u_1 = -u_2 \in U_1 \cap U_2 = \{0\}$, so $u_1 = u_2 = 0$.

(ii) In $\mathbf{F}^2$, take $U_1 = \operatorname{span}(e_1)$, $U_2 = \operatorname{span}(e_2)$, $U_3 = \operatorname{span}(e_1 + e_2)$. Each pairwise intersection is $\{0\}$. But $(e_1 + e_2) = e_1 + e_2 + 0 = 0 + e_2 + e_1$, giving two distinct representations of $e_1 + e_2$ in $U_1 + U_2 + U_3$. The Condition for a Direct Sum fails: $e_1 + e_2 + (-e_1 - e_2) = 0$ has the non-trivial decomposition $e_1 \in U_1$, $e_2 \in U_2$, $-e_1 - e_2 \in U_3$.

Part (c): The zero vector is absorbing.

(i) $0v = (0 + 0)v = 0v + 0v$ by axiom (viii). Adding $-(0v)$ to both sides (which exists by axiom (v)) gives $0 = 0v$.

(ii) $a0 = a(0 + 0) = a0 + a0$ by axiom (vii). Adding $-(a0)$ gives $0 = a0$.

(iii) $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$ by axiom (viii) and (i). By uniqueness of additive inverses, $(-1)v = -v$.

(iv) $(-1)(-1)v = (-1)(-v) = -(-v)$ by (iii) applied twice. By uniqueness of the additive inverse, $-(-v) = v$. Therefore $(-1)(-1)v = v$. This uses axiom (ix) (associativity of scalar multiplication) implicitly via the identification $(-1)(-1) = 1$.

Part (d): The subspace lattice.

(i) $U_1 \cap U_2$ is non-empty (both contain 0). If $u, v \in U_1 \cap U_2$, then $u, v \in U_1$ so $u + v \in U_1$, and $u, v \in U_2$ so $u + v \in U_2$; thus $u + v \in U_1 \cap U_2$. Scalar closure follows similarly. By the three-conditions criterion, $U_1 \cap U_2$ is a subspace.

(ii) ($\Leftarrow$) If $U_1 \subseteq U_2$, then $U_1 \cup U_2 = U_2$ is a subspace. Symmetrically for $U_2 \subseteq U_1$. ($\Rightarrow$) Suppose $U_1 \not\subseteq U_2$ and $U_2 \not\subseteq U_1$. Pick $u \in U_1 \setminus U_2$ and $w \in U_2 \setminus U_1$. Then $u + w \in U_1 \cup U_2$ (if it were in $U_1 \cup U_2$, it would lie in U_1 or U_2; in U_1: $w = (u + w) - u \in U_1$, contradiction; in U_2: $u = (u + w) - w \in U_2$, contradiction). So $U_1 \cup U_2$ is not closed under addition, hence not a subspace.

(iii) $\cap$ is associative and commutative (set operations); the meet of U_1 and U_2 is $U_1 \cap U_2$, which is a subspace by (i). The join is $U_1 + U_2$, which is a subspace by the Sum of Subspaces theorem and is the smallest subspace containing both. Absorption laws hold: $U_1 \cap (U_1 + U_2) = U_1$ and $U_1 + (U_1 \cap U_2) = U_1$ follow from $U_1 \subseteq U_1 + U_2$ and $U_1 \cap U_2 \subseteq U_1$. The lattice is therefore verified.

Part (e): Decomposition of the tangent space for the torus simulation.

(i) $D = \{(v, v, \ldots, v) : v \in \mathbb{R}^2\} \cong \mathbb{R}^2$, so $\dim D = 2$. It is a subspace: $0 \in D$, closed under coordinatewise addition $(v, \ldots, v) + (w, \ldots, w) = (v + w, \ldots, v + w)$, and closed under scalar multiplication.

(ii) $R = \{(v_1, \ldots, v_n) : v_1 + \cdots + v_n = 0\}$. This is a subspace (the kernel of the linear map $\Sigma : (\mathbb{R}^2)^n \to \mathbb{R}^2$, $(v_1, \ldots, v_n) \mapsto v_1 + \cdots + v_n$). Its dimension is $2n-2$ (the kernel of a surjection from a $2n$-dimensional space to a 2-dimensional space).

(iii) $D + R = (\mathbb{R}^2)^n$: given any $(v_1, \ldots, v_n)$, let $\bar{v} = \frac{1}{n}(v_1 + \cdots + v_n)$ and write $(v_1, \ldots, v_n) = (\bar{v}, \ldots, \bar{v}) + (v_1 - \bar{v}, \ldots, v_n - \bar{v})$; the first term is in D and the second is in R (its components sum to zero). $D \cap R = \{0\}$: if $(v, \ldots, v) \in R$ then $nv = 0$ so $v = 0$. By the Direct Sum of Two Subspaces theorem, $D \cap R = \{0\}$ and $D + R = (\mathbb{R}^2)^n$ give $(\mathbb{R}^2)^n = D \oplus R$. □

Remark (Dependencies).

- Definition: Vector Space
- Definition: Subspace
- Theorem: Conditions for a Subspace
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$
- Theorem: $a0 = 0$

Chapter 4

Lattice Order

lattice-order

Sets, Relations, and Functions → **Lattice Order** → Algebras of Sets

Structural Roadmap

How we got here. The preceding chapter on sets, relations, and functions already introduced partial orders, chains, suprema, infima, and the first appearance of lattices as an example of order structure. The material there was sufficient for the purposes of that chapter.

What this chapter develops. Here order is the main object of study rather than an incidental feature. The chapter builds a self-contained theory of ordered sets: the canonical definitions of partial, strict, and total orders; chains and antichains; maps between ordered sets; the duality principle; and the covering relation and its role in finite posets.

Where this leads. Lattice order is the structural substrate beneath algebras of sets, sigma-algebras, topologies, Boolean algebras, and fixed-point theorems. The fuller development of distributive, modular, and complete lattices belongs in a later chapter dedicated to that theory.

4.0.1 Why Lattice Theory Gets Its Own Home

Lattice theory begins with a simple observation: in many ordered structures, two elements have a natural least upper bound and a natural greatest lower bound. Once those operations exist everywhere, order starts behaving like algebra.

Remark 4.0.1 (Where the first pass already lives). The first introduction to lattices is already in the lattice section of the order notes. That is the right place for the first encounter, because it grows directly out of partial orders, suprema, and infima.

Remark 4.0.2 (Why return later). Once lattices stop being examples and start becoming the main character, the story gets richer. A fuller lattice-theory chapter will naturally gather topics such as:

- distributive, modular, and Boolean lattices;

- complete lattices and closure operators;
- sublattices, ideals, and filters;
- Galois connections and adjoint situations;
- fixed-point theorems such as Knaster-Tarski;
- the lattice viewpoint behind topologies, sigma-algebras, and domain theory.

Remark 4.0.3 (Why this matters for the rest of the curriculum). This chapter is a breadcrumb because lattice order quietly reappears all over the place. Power sets form Boolean lattices, sigma-algebras behave like closure-stable lattice structures, open sets form a join-rich order, and fixed-point methods show up in logic, measure, computation, and analysis. Giving lattice order its own future home makes those later echoes easier to recognize.

4.1 Ordered Sets

Toolkit: Ordered Sets

Order is a relation, not a property. A set carries no intrinsic ordering; an order is an additional structure imposed on the set by specifying which pairs of elements stand in the ordering relation. The same underlying set can support many distinct orders simultaneously. The atomic notions developed here are:

- the **partial order** the minimal framework in which order makes sense;
- the **strict partial order** the asymmetric companion to the partial order, and the native language of open-ball conditions and limits;
- the **total order** the specialisation in which every pair of elements is comparable.

Definition (Partial Order)

Definition 4.1.1 (Partial Order). A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \Longrightarrow a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \Longrightarrow a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Remark (Standard quantified statement).

$$\mathrm{PartialOrder}(S, \leq) \;:=\; \big(\forall a \in S,\ a \leq a\big) \wedge \big(\forall a, b \in S,\ a \leq b \wedge b \leq a \Rightarrow a = b\big) \wedge \big(\forall a, b, c \in S,\ a \leq b \wedge b \leq$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{PartialOrder}(S, \leq) \;:=\; \big(\exists a \in S,\ a \not\leq a\big) \vee \big(\exists a, b \in S,\ a \leq b \wedge b \leq a \wedge a \neq b\big) \vee \big(\exists a, b, c \in S,\ a \leq b \wedge b \leq$$

Remark (Interpretation). Order is not an intrinsic quality of a set's elements it is an external binary relation. The three axioms together enforce that the relation be a consistent, cycle-free, directed comparison. Reflexivity says every element is at least as large as itself. Anti-symmetry is the formal engine behind squeeze arguments: if $a \leq b$ and $b \leq a$ then a and b cannot be distinct. Transitivity ensures that the ordering chains: comparisons can be composed without breaking consistency. The same set of objects can carry many distinct partial orders simultaneously standard, reverse, or lexicographic each defining a different relational geometry.

Definition 4.1.2 (Strict Partial Order). A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \implies \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \implies a < c$.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(S, <) \;:=\; \big(\forall a \in S,\ \neg(a < a)\big) \wedge \big(\forall a, b \in S,\ a < b \Rightarrow \neg(b < a)\big) \wedge \big(\forall a, b, c \in S,\ a <$$

Remark (Interpretation). The strict order is the companion of the non-strict order, eliminating reflexivity and replacing anti-symmetry with the stronger asymmetry. Every strict partial order determines a partial order by $a \leq b \iff (a < b \vee a = b)$, and every partial order determines a strict partial order by $a < b \iff (a \leq b \wedge a \neq b)$. In metric spaces and analysis, strict orders are the natural language of open conditions: open balls, open intervals, and limit definitions all use $<$ rather than $\leq$ to exclude boundary contact.

Remark (Dependencies).

- Partial Order

Definition (Total Order)

Definition 4.1.3 (Total Order). A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

Remark (Standard quantified statement).

$$\mathrm{TotalOrder}(S, \leq) \;:=\; \mathrm{PartialOrder}(S, \leq) \;\wedge\; \forall a, b \in S,\ (a < b) \oplus (a = b) \oplus (b < a),$$

where $\oplus$ denotes exclusive or. Equivalently, comparability suffices: $\forall a, b \in S,\ a \leq b \vee b \leq a$.

Remark (Interpretation). A partial order permits elements to be incomparable: it is possible that neither $a \leq b$ nor $b \leq a$. A total order removes this possibility entirely every pair of elements is forced to be comparable. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ under their standard orderings are totally ordered fields. By contrast, the power set $\mathcal{P}(X)$ of any set X with more than one element, ordered by inclusion, is typically not a total order: the singletons $\{a\}$ and $\{b\}$ with $a \neq b$ are incomparable.

Remark (Dependencies). Partial order, strict partial order.

4.2 Chains and Antichains

Toolkit: Chains and Antichains

A poset can contain subsets whose elements are either completely aligned or entirely incomparable. These two extremes are formalised as chains and antichains. They measure the "linearity" and "width" of the order structure respectively.

Definition (Chain)

Definition 4.2.1 (Chain). Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Remark (Standard quantified statement).

$$\mathrm{Chain}(C, S, \leq) \ :=\ C \subseteq S \ \wedge \ \forall x, y \in C,\ x \leq y \vee y \leq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Chain}(C, S, \leq) \ :=\ \exists x, y \in C,\ x \not\leq y \wedge y \not\leq x.$$

Remark (Interpretation). A chain is a subset in which the partial order behaves as a total order: no two elements are left incomparable. The entire real line $\mathbb{R}$ under its standard ordering is itself a chain. Any finite totally ordered set a ranking, a sequence of nested intervals, a linearly ordered list of stages is a chain. Chains model the "ladder" structure of order: there is a clear direction from smaller to larger, and the path from any element to any other is unambiguous.

Remark (Dependencies). Partial order, total order.

Definition (Antichain)

Definition 4.2.2 (Antichain). Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

Remark (Standard quantified statement).

$$\mathrm{Antichain}(A, S, \leq) \ :=\ A \subseteq S \ \wedge \ \forall x, y \in A,\ x \neq y \Rightarrow x \not\leq y \wedge y \not\leq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Antichain}(A, S, \leq) \;\coloneqq\; \exists x, y \in A,\; x \neq y \wedge (x \leq y \vee y \leq x).$$

Remark (Interpretation). An antichain is a subset in which the partial order makes no comparisons at all: every pair of distinct elements is incomparable. In the power set $\mathcal{P}(X)$ ordered by inclusion, the collection of all subsets of a fixed cardinality k forms an antichain no subset of size k contains another. Antichains measure the "width" of a poset: Dilworth's theorem states that the minimum number of chains needed to cover a finite poset equals the maximum size of an antichain. In higher-dimensional structures such as product orders, large antichains arise naturally and are the primary tool for understanding the lateral spread of the order.

Remark (Dependencies). Partial order.

4.3 Partial and Total Maps

Toolkit: Partial and Total Maps

Before studying maps that preserve order structure, the foundational function types are recorded: a partial map need not be defined everywhere, while a total map is defined on every input. These are the base layer upon which the order-theoretic notions rest.

Definition (Partial Map)

Definition 4.3.1 (Partial Map). Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\; \forall y_1, y_2 \in B, \quad (x, y_1) \in f \;\wedge\; (x, y_2) \in f \implies y_1 = y_2.$$

Remark (Standard quantified statement).

$$\mathrm{PartialMap}(f, A, B) \;\coloneqq\; f \subseteq A \times B \wedge \forall x \in A,\; \forall y_1, y_2 \in B,\; (x, y_1) \in f \wedge (x, y_2) \in f \Rightarrow y_1 = y_2.$$

Remark (Interpretation). A partial map assigns to each element of A at most one element of B. Its domain of definition $\mathrm{dom}(f) = \{x \in A : \exists y \in B,\, (x, y) \in f\}$ may be a proper subset of A. Partial maps arise in order theory when a function is well-defined only on elements satisfying some order-theoretic condition for example, on elements that admit an upper bound.

Definition 4.3.2 (Total Map). A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\; \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

Remark (Standard quantified statement).

$$\mathrm{TotalMap}(f, A, B) \;\coloneqq\; \mathrm{PartialMap}(f, A, B) \;\wedge\; \forall x \in A,\; \exists y \in B,\; (x, y) \in f.$$

Remark (Dependencies). Partial map.

4.4 Maps Between Ordered Sets

Toolkit: Maps Between Ordered Sets

The structure-preserving maps between posets form a hierarchy of increasing strength. An order-preserving map carries the direction of the order forward. An order-embedding carries it in both directions. An order-isomorphism is a surjective order-embedding a perfect structural identification between two posets.

Definition (Order-Preserving Map)

Definition 4.4.1 (Order-Preserving Map). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Remark (Standard quantified statement).

$$\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\ x \leq_P y \Rightarrow \varphi(x) \leq_Q \varphi(y).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \exists x, y \in P,\ x \leq_P y \wedge \varphi(x) \not\leq_Q \varphi(y).$$

Remark (Interpretation). An order-preserving map carries the direction of the order from P into Q: whenever x is at or below y in P, $\varphi(x)$ is at or below $\varphi(y)$ in Q. The implication is one-directional order-preserving maps need not reflect the order back. Two order-preserving maps compose to an order-preserving map, making order-preserving maps the morphisms of the category of posets.

Remark (Dependencies). Partial order.

Definition (Order-Embedding)

Definition 4.4.2 (Order-Embedding). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\ x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

Remark (Interpretation). An order-embedding both preserves and reflects the order: the order structure of P is faithfully copied into Q without distortion. Unlike mere preservation, reflection means that order-relations between images in Q genuinely originate from order-relations in P. Every order-embedding is injective (Lemma 4.4.4) and identifies P order-isomorphically with its image in Q.

Remark (Dependencies). Order-preserving map.

Definition (Order-Isomorphism)

Definition 4.4.3 (Order-Isomorphism). An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \;:=\; \mathrm{OrderEmbedding}(\varphi, P, Q) \;\wedge\; \forall q \in Q,\; \exists p \in P,\; \varphi(p) = q.$$

Remark (Interpretation). An order-isomorphism is a bijection that perfectly translates the order of P into Q and back. Two order-isomorphic posets are order-theoretically identical: any statement about P expressed purely in terms of $\leq_P$ translates to an equally valid statement about Q under φ. Order-isomorphism is the correct notion of equality for posets.

Remark (Dependencies). Order-embedding.

Lemma 4.4.4 (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.*

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \implies \forall x, y \in P,\; \varphi(x) = \varphi(y) \Rightarrow x = y.$$

Remark (Interpretation). Injectivity follows automatically from the biconditional in the embedding definition. If $\varphi(x) = \varphi(y)$, the image is simultaneously $\leq$ and $\geq$ itself in Q; order-reflection pulls this back to P, and anti-symmetry then forces $x = y$.

Remark (Dependencies). Order-embedding, anti-symmetry of partial orders.

Remark (Proof). Go to proof.

4.5 The Covering Relation

Toolkit: The Covering Relation

The covering relation identifies the irreducible steps of a partial order: y covers x when $x < y$ and there is nothing strictly between them. In a finite poset the entire order is determined by its covering relation, which is encoded exactly by the Hasse diagram.

Definition (Covering Relation)

Definition 4.5.1 (Covering Relation). Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \;:=\; x < y \;\wedge\; \neg\big(\exists z \in P,\; x < z < y\big).$$

Remark (Standard quantified statement).

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \forall z \in P,\; x < z \leq y \Longrightarrow z = y.$$

Remark (Negated quantified statement).

$$\neg(x \prec y) \;\coloneqq\; x \not< y \;\vee\; \exists z \in P,\; x < z < y.$$

Remark (Interpretation). The covering relation captures the direct steps of the order the comparisons that cannot be decomposed further. In a finite poset every strict comparison $x < y$ can be resolved into a finite chain of covers $x = x_0 \prec x_1 \prec \cdots \prec x_n = y$, so the covering relation alone determines the full order. The Hasse diagram of a finite poset is exactly the directed graph of the covering relation. In a dense order such as $\mathbb{Q}$ or $\mathbb{R}$, no element covers any other: between any two elements there is always a third.

Remark (Dependencies). Partial order, strict partial order.

Lemma 4.5.2 (Characterizations of Order-Isomorphisms for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$:* $\quad x <_P y \iff \varphi(x) <_Q \varphi(y)$.
3. *For all $x, y \in P$:* $\quad x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \iff \big(\forall x, y \in P,\; x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)\big) \iff \big(\forall x, y \in P,\; x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi($$

Remark (Interpretation). In a finite poset the order relation, the strict order, and the covering relation each determine the others completely. This lemma shows that an order-isomorphism can be detected at any one of these three levels. The covering characterisation is the most useful in practice: since Hasse diagrams encode exactly the covering relation, two finite posets are order-isomorphic precisely when their diagrams are isomorphic as directed graphs.

Remark (Dependencies). Order-isomorphism, covering relation.

Remark (Proof). Go to proof.

Proposition 4.5.3 (Hasse Diagram Characterization for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs that is, identical up to relabeling of elements.*

Remark (Standard quantified statement).

$$P \cong Q \iff \exists \varphi : P \xrightarrow{\sim} Q,\; \forall x, y \in P,\; x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y).$$

Remark (Interpretation). The Hasse diagram is a complete invariant of the order structure of a finite poset. Checking order-isomorphism between finite posets reduces to a graph isomorphism problem on the covering graphs two finite posets with non-isomorphic Hasse diagrams cannot be order-isomorphic.

Remark (Dependencies). Characterizations of order-isomorphisms for finite posets, covering relation.

Remark (Proof). Go to proof.

4.6 Duality

Toolkit: Duality

Every partially ordered set has a dual obtained by reversing all order relations. Any statement that holds in every poset therefore holds in every dual poset, which means its order-reversed version also holds universally. This is the Duality Principle the formal engine behind phrases like "by duality, the corresponding result for infima follows."

Definition (Dual Ordered Set)

Definition 4.6.1 (Dual Ordered Set). Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Remark (Standard quantified statement).

$$x \leq_{P^{\mathrm{op}}} y \coloneqq y \leq_P x.$$

Remark (Interpretation). Dualising a poset exchanges the roles of upper and lower: every upper bound becomes a lower bound, every supremum becomes an infimum, and every maximal element becomes a minimal element. In a finite poset the Hasse diagram of P^{op} is obtained from that of P by turning it upside down. The dual of a total order is again a total order.

Remark (Dependencies). Partial order.

Proposition 4.6.2 (Duality Principle). *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.*

Remark (Interpretation). The Duality Principle eliminates the need to re-prove results whose dual is already established. Any theorem about upper bounds, suprema, maximal elements, or top elements automatically yields a theorem about lower bounds, infima, minimal elements, or bottom elements by applying the principle to P^{op}.

Remark (Dependencies). Dual ordered set.

Remark (Proof). Go to proof.

4.7 Extremal Elements

Toolkit: Extremal Elements

A poset may contain elements that sit at the boundary of the order. There are two distinct senses of "extreme": an element can be extreme *locally* (nothing in the poset exceeds it, though it may be incomparable with some elements) or extreme *globally* (it dominates or is dominated by every element). The definitions below make these distinctions precise, proceeding from global extremes of the whole poset, through local extremes of a subset, to the proposition that finite posets always contain the local kind.

Definition (Bottom Element)

Definition 4.7.1 (Bottom Element). Let $(P, \leq)$ be a partially ordered set. An element $\perp \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \perp \leq x.$$

If a bottom element exists it is unique; it is denoted $\perp_P$ or simply $\perp$.

Remark (Standard quantified statement).

$$\mathrm{Bottom}(\perp, P, \leq) \;:=\; \perp \in P \;\wedge\; \forall x \in P,\ \perp \leq x.$$

Remark (Interpretation). A bottom element is below every element of the poset without exception. In $(\mathcal{P}(X), \subseteq)$ the bottom element is $\varnothing$. The natural numbers have a bottom element; the integers do not. An antichain with more than one element has no bottom element, since no member is below any other.

Remark (Dependencies). Partial order.

Definition (Top Element)

Definition 4.7.2 (Top Element). Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Remark (Standard quantified statement).

$$\mathrm{Top}(\top, P, \leq) \;:=\; \top \in P \;\wedge\; \forall x \in P,\ x \leq \top.$$

Remark (Interpretation). Dual to the bottom element via the Duality Principle. In $(\mathcal{P}(X), \subseteq)$ the top element is X itself.

Remark (Dependencies). Bottom element, duality principle.

Definition 4.7.3 (Maximal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Remark (Standard quantified statement).

$$\mathrm{Maximal}(a, Q, \leq) \;:=\; a \in Q \;\wedge\; \forall x \in Q,\; a \leq x \Rightarrow a = x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Maximal}(a, Q, \leq) \;:=\; \exists x \in Q,\; a < x,$$

that is, some element of Q strictly exceeds a.

Remark (Interpretation). A maximal element asserts only that nothing in Q strictly dominates it. It need not be comparable with every element of Q, so a poset can have multiple maximal elements simultaneously. In domain theory, maximal elements correspond to fully-specified or total objects.

Remark (Dependencies). Partial order.

Definition 4.7.4 (Minimal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Remark (Standard quantified statement).

$$\mathrm{Minimal}(a, Q, \leq) \;:=\; a \in Q \;\wedge\; \forall x \in Q,\; x \leq a \Rightarrow x = a.$$

Remark (Interpretation). Dual to maximal element via the Duality Principle.

Remark (Dependencies). Maximal element, duality principle.

Definition 4.7.5 (Greatest Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Remark (Standard quantified statement).

$$\mathrm{Greatest}(m, Q, \leq) \;:=\; m \in Q \;\wedge\; \forall x \in Q,\; x \leq m.$$

Remark (Interpretation). A greatest element is a maximal element that is additionally comparable with every element of Q. When $\max Q$ exists, $\mathrm{Max}(Q) = \{\max Q\}$, but the converse fails: a unique maximal element need not dominate every element.

Remark (Dependencies). Maximal element.

Definition 4.7.6 (Least Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Remark (Standard quantified statement).

$$\mathrm{Least}(m, Q, \leq) \;:=\; m \in Q \;\wedge\; \forall x \in Q,\; m \leq x.$$

Remark (Interpretation). Dual to greatest element.

Remark (Dependencies). Greatest element, duality principle.

Proposition 4.7.7 (Every Nonempty Finite Poset Has a Maximal Element). *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.*

Remark (Standard quantified statement).

$$P \text{ finite},\; Q \subseteq P,\; Q \neq \varnothing \implies \mathrm{Max}(Q) \neq \varnothing.$$

$$\forall x \in P,\; \exists y \in \mathrm{Max}(P),\; x \leq y.$$

Remark (Interpretation). Finiteness is essential. The collection of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element. In a finite poset one can always ascend from any element to a ceiling. This property is the finite shadow of Zorn's Lemma and underlies ascending chain arguments throughout algebra and order theory.

Remark (Dependencies). Maximal element, partial order.

Remark (Proof). Go to proof.

4.8 Lifting and Co-Lifting

Toolkit: Lifting and Co-Lifting

A poset that lacks a bottom or top element can be given one by adjoining a new universal bound. Lifting adjoins a bottom; co-lifting adjoins a top. These constructions are dual to each other and are used throughout domain theory, where a bottom element represents an undefined state.

Definition (Lifting)

Definition 4.8.1 (Lifting). Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Remark (Standard quantified statement).

$$P_\perp = P \cup \{\perp\}, \quad \perp \notin P, \quad \forall x \in P_\perp,\; \perp \leq x, \quad \forall x, y \in P,\; x \leq_{P_\perp} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Lifting adjoins a universal lower bound to a poset that may not have one. In domain theory $\perp$ represents an undefined, divergent, or unspecified state. Any set S viewed as an antichain can be lifted to the *flat poset* $S_\perp$, in which all elements of S are incomparable above $\perp$.

Remark (Dependencies). Partial order, bottom element.

Definition 4.8.2 (Co-Lifting). Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

Remark (Standard quantified statement).

$$P^\top = P \cup \{\top\}, \quad \top \notin P, \quad \forall x \in P^\top,\ x \leq \top, \quad \forall x, y \in P,\ x \leq_{P^\top} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Dual to lifting via the Duality Principle: co-lifting adjoins a universal upper bound. The co-lifting of an antichain produces a flat poset in which all elements are incomparable below $\top$.

Remark (Dependencies). Lifting, top element, duality principle.

4.9 Consistency

Toolkit: Consistency

Two elements of a poset are consistent when they share a common upper bound when they can be jointly extended to some element above both of them. In lattice-theoretic settings this is automatic; in general posets it is a genuine condition.

Definition 4.9.1 (Consistency). Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge\ y \leq z.$$

Remark (Standard quantified statement).

$$\text{Consistent}(x, y, S, \leq) \ :=\ \exists z \in S,\ x \leq z \wedge y \leq z.$$

Remark (Negated quantified statement).

$$\neg\,\text{Consistent}(x, y, S, \leq) \ :=\ \forall z \in S,\ x \nleq z \vee y \nleq z.$$

Remark (Interpretation). Consistency asserts that x and y are compatible there is some element of S above both. In a lattice where binary joins always exist, consistency is automatic: take $z = x \vee y$. In a general poset the condition is nontrivial and may fail. In domain theory, consistency captures the idea that two partial computations describe information that can be combined without contradiction.

Remark (Dependencies). Partial order.

4.10 Down-Sets and Up-Sets

Toolkit: Down-Sets and Up-Sets

A down-set is a subset of a poset that is closed downward: if an element belongs to it, everything below that element belongs to it too. An up-set is the dual notion closed upward. Down-sets are the order ideals of lattice theory; up-sets are the order filters. The collection of all down-sets of a poset, ordered by inclusion, is itself an important poset.

Definition (Down-Set)

Definition 4.10.1 (Down-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\ \forall y \in P, \quad y \leq x \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{DownSet}(Q, P, \leq) \;\coloneqq\; Q \subseteq P \;\wedge\; \forall x \in Q,\ \forall y \in P,\ y \leq x \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DownSet}(Q, P, \leq) \;\coloneqq\; \exists x \in Q,\ \exists y \in P,\ y \leq x \wedge y \notin Q.$$

Remark (Interpretation). A down-set is a downward-closed region of the poset: once an element is included, the entire lower section below it is forced in. Down-sets generalise downward-closed intervals. The collection of all down-sets of P, denoted $\mathcal{O}(P)$ and ordered by inclusion, is itself a poset and in the finite case a distributive lattice. Birkhoff's representation theorem states that every finite distributive lattice arises this way.

Remark (Dependencies). Partial order.

Definition (Up-Set)

Definition 4.10.2 (Up-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\ \forall y \in P, \quad x \leq y \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{UpSet}(Q, P, \leq) \;\coloneqq\; Q \subseteq P \;\wedge\; \forall x \in Q,\ \forall y \in P,\ x \leq y \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{UpSet}(Q, P, \leq) \;\coloneqq\; \exists x \in Q,\ \exists y \in P,\ x \leq y \wedge y \notin Q.$$

Remark (Interpretation). Dual to down-set via the Duality Principle: the up-sets of P are exactly the down-sets of P^{op}. In topology, filters are special up-sets closed under finite intersection and satisfying a directedness condition. The complement of a down-set is an up-set, and vice versa.

Remark (Dependencies). Down-set, duality principle.

4.11 Families of Sets

Toolkit: Families of Sets

The power set of any set is naturally ordered by inclusion. Every collection of subsets whether arising from algebra, topology, or logic inherits this order. The key observation is that subsets correspond exactly to predicates via their characteristic functions, and that this correspondence is an order-isomorphism.

Let X be a set. The *power set* $\mathcal{P}(X)$, consisting of all subsets of X, is ordered by inclusion: for $A, B \in \mathcal{P}(X)$,

$$A \leq B \iff A \subseteq B.$$

Any subcollection of $\mathcal{P}(X)$ inherits this order. Such a collection is called a *family of sets*.

Families of sets arise naturally when X carries additional structure. If X is a group G, the set $\mathrm{Sub}(G)$ of all subgroups and the set $\mathrm{NSub}(G)$ of all normal subgroups are each ordered by inclusion. If X is a vector space V, the set $\mathrm{Sub}(V)$ of all subspaces is ordered by inclusion. If X is a ring R, the sets of all subrings and of all ideals of R are similarly ordered. If $(X, \mathcal{T})$ is a topological space, the collections of open sets, closed sets, and clopen sets are each families of sets ordered by inclusion.

A particularly important example is the family $\mathcal{O}(P)$ of all down-sets of a partially ordered set P, ordered by inclusion.

The ordered set $(\mathcal{P}(X), \subseteq)$ admits an equivalent formulation in terms of predicates. A *predicate* on X is a function

$$p : X \to \{T, F\},$$

where T and F denote truth values. Let $\mathcal{P}^*(X)$ denote the set of all predicates on X, ordered by implication:

$$p \leq q \iff \{x \in X : p(x) = T\} \subseteq \{x \in X : q(x) = T\}.$$

Define the map

$$\Phi : \mathcal{P}^*(X) \to \mathcal{P}(X), \qquad \Phi(p) := \{x \in X : p(x) = T\}.$$

Proposition 4.11.1 (Predicates and Power Sets Are Order-Isomorphic). *The map $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ is an order-isomorphism.*

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\Phi, \mathcal{P}^*(X), \mathcal{P}(X)).$$

Remark (Interpretation). Every subset of X can be represented as the truth set of a predicate, and every predicate determines a subset. The correspondence is bijective and order-preserving in both directions: $p \leq q$ if and only if the truth set of p is contained in the truth set of q, which is exactly the inclusion order on $\mathcal{P}(X)$. This identification is fundamental in logic, where sets and their characteristic predicates are treated as interchangeable, and in computer science, where sets are often implemented via their membership tests.

Remark (Dependencies). Order-isomorphism, partial order.

Proofs

Proofs

Remark (Return). Return to Lemma 4.4.4.

Theorem (Every order-embedding is injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.*

Proof. Suppose $\varphi(x) = \varphi(y)$. Then

$$\varphi(x) \leq_Q \varphi(y) \quad \text{and} \quad \varphi(y) \leq_Q \varphi(x).$$

Since φ is an order-embedding, the biconditional $x \leq_P y \iff \varphi(x) \leq_Q \varphi(y)$ holds for all $x, y \in P$. Order-reflection therefore gives

$$x \leq_P y \quad \text{and} \quad y \leq_P x.$$

By anti-symmetry of $\leq_P$, it follows that $x = y$. □

Remark (Proof structure). A direct proof by chasing the biconditional in the definition of order-embedding. The only content is anti-symmetry, applied once.

Remark (Dependencies). Order-embedding, anti-symmetry of partial orders.

Remark (Return). Return to Lemma 4.5.2.

Theorem (Characterizations of Order-Isomorphisms for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent: (i) φ is an order-isomorphism; (ii) $x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)$ for all $x, y \in P$; (iii) $x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y)$ for all $x, y \in P$.*

Proof. **(i) $\Leftrightarrow$ (ii).** The equivalence follows directly from the definitions: φ is an order-isomorphism if and only if $x \leq_P y \Leftrightarrow \varphi(x) \leq_Q \varphi(y)$. Since φ is bijective, $x = y$ if and only if $\varphi(x) = \varphi(y)$, so $x <_P y$ ($x \leq_P y$ and $x \neq y$) if and only if $\varphi(x) <_Q \varphi(y)$.

(ii) $\Rightarrow$ (iii). Assume (ii). Suppose $x \prec_P y$, so $x <_P y$ and there is no z with $x <_P z <_P y$. By (ii), $\varphi(x) <_Q \varphi(y)$. If there existed $w \in Q$ with $\varphi(x) <_Q w <_Q \varphi(y)$, then since φ is bijective, $w = \varphi(u)$ for some $u \in P$, and (ii) gives $x <_P u <_P y$, contradicting $x \prec_P y$. Therefore $\varphi(x) \prec_Q \varphi(y)$. The reverse implication is symmetric.

(iii) $\Rightarrow$ (ii). Assume (iii). Suppose $x <_P y$. Since P is finite, there exists a covering chain $x = x_0 \prec_P x_1 \prec_P \cdots \prec_P x_n = y$. By (iii), $\varphi(x_0) \prec_Q \cdots \prec_Q \varphi(x_n)$, so $\varphi(x) <_Q \varphi(y)$. The reverse implication follows by bijectivity of φ. □

Remark (Proof structure). A three-way cycle of implications. The key step is (ii) $\Rightarrow$ (iii): a covering is an irreducible strict comparison, and if a covering is reflected by a covering then there can be no intermediate element in Q (by bijectivity carrying it back to P).

Remark (Dependencies). Order-isomorphism, covering relation, finiteness of P (used in (iii) $\Rightarrow$ (ii) to decompose $<$ into a covering chain).

Remark (Return). Return to Proposition 4.5.3.

Theorem (Hasse Diagram Characterization for Finite Posets). *Finite partially ordered sets P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs.*

Proof. ($\Rightarrow$). If $\varphi : P \to Q$ is an order-isomorphism, then by Lemma 4.5.2, φ preserves and reflects the covering relation. Therefore φ induces a bijection between the edges of the Hasse diagram of P and those of the Hasse diagram of Q that respects the direction of each edge. The Hasse diagrams are thus isomorphic as directed graphs.

($\Leftarrow$). Suppose there exists a bijection $\varphi : P \to Q$ that is an isomorphism of the Hasse diagrams, so $x \prec_P y$ if and only if $\varphi(x) \prec_Q \varphi(y)$. By Lemma 4.5.2, condition (iii), φ is an order-isomorphism. □

Remark (Proof structure). An immediate consequence of the covering characterisation of order-isomorphisms for finite posets, applied in both directions. The covering relation is precisely the edge relation of the Hasse diagram, so isomorphism of the diagrams is equivalent to order-isomorphism.

Remark (Dependencies). Characterizations of order-isomorphisms for finite posets, covering relation, order-isomorphism.

Remark (Return). Return to Proposition 4.7.7.

Theorem (Every nonempty finite poset has a maximal element). *Let* $(P, \leq)$ *be a finite partially ordered set and let* $Q \subseteq P$ *be nonempty. Then* $\mathrm{Max}(Q) \neq \varnothing$.

Proof. Since Q is nonempty, pick any $a_0 \in Q$. If a_0 is maximal in Q, we are done. Otherwise there exists $a_1 \in Q$ with $a_0 < a_1$. Continuing, if a_1 is not maximal there exists $a_2 \in Q$ with $a_1 < a_2$. The sequence $a_0 < a_1 < a_2 < \cdots$ is strictly increasing in Q. Since $Q \subseteq P$ and P is finite, this sequence cannot continue indefinitely. Therefore it must terminate at some $a_n \in Q$ from which no further strict increase is possible, and a_n is maximal in Q.

For the second part, given any $x \in P$, apply the above argument to $Q = \{y \in P : x \leq y\}$, which is nonempty (it contains x). A maximal element of Q is an element of $\mathrm{Max}(P)$ that is above x. $\square$

Remark (Proof structure). An ascending chain argument. The chain $a_0 < a_1 < \cdots$ must terminate because P is finite. Finiteness is essential: the set of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element.

Remark (Dependencies). Maximal element, partial order (strict form).

Remark (Return). Return to Proposition 4.6.2.

Theorem (Duality Principle). *Let Φ be a statement about partially ordered sets expressed in terms of $\leq$ that is true in every partially ordered set. Then the dual statement Φ^{op}, obtained by reversing the order, is also true in every partially ordered set.*

Proof. Let P be any partially ordered set. The dual P^{op} is also a partially ordered set (Definition 4.6.1). Since Φ holds in every partially ordered set, Φ holds in P^{op}. The statement Φ applied to P^{op} is, by definition of P^{op}, precisely the statement Φ^{op} applied to P. Therefore Φ^{op} holds in P. Since P was arbitrary, Φ^{op} holds in every partially ordered set. □

Remark (Proof structure). A one-step argument: the dual P^{op} is a valid poset, so any universal statement about posets applies to it. Reading that statement back in the original order of P yields the dual statement.

Remark (Dependencies). Dual ordered set, partial order.

Chapter 5

Boolean Algebra

Chapter Status

This chapter is planned. It will collect Boolean algebra material and related set-algebraic structure. The chapter is currently a placeholder for future notes based on Boolean algebra study.

Planned Scope

- Boolean algebras
- Boolean operations
- Boolean identities
- Boolean rings, if appropriate
- Power-set Boolean algebra
- Connections with propositional logic
- Connections with set operations

Notes

These definitions and notes will be added in a future rebuild pass.

Proofs

Proofs

Proof entries for this chapter are planned.

Exercises

Exercises

Exercise entries for this chapter are planned.

Bibliography